Setting up AI Projects

AI Projects
Car Plate Recognition

AI Projects

WHEN A USER TAKES A PHOTO, THE APP SHOULD CHECK WHETHER THEY'RE IN A NATIONAL PARK...
SURE, EASY GIS LOOKUP. GIMME A FEW HOURS.
...AND CHECK WHETHER THE PHOTO IS OF A BIRD.
I'LL NEED A RESEARCH TEAM AND FIVE YEARS.
IN CS, IT CAN BE HARD TO EXPLAIN THE DIFFERENCE BETWEEN THE EASY AND THE VIRTUALLY IMPOSSIBLE.

Machine Learning Projects

85% of AI projects fail[1]

1 Pactera Technologies

Why do so many projects fail?

- ML is still research - you shouldn’t aim for 100% success rate
- But, many are doomed to fail:
  - Technically infeasible or poorly scoped

Why do so many projects fail?

- ML is still research - you shouldn’t aim for 100% success rate
- But, many are doomed to fail:
  - Technically infeasible or poorly scoped
  - Never make the leap to production

Why do so many projects fail?

- ML is still research - you shouldn’t aim for 100% success rate
- But, many are doomed to fail:
  - Technically infeasible or poorly scoped
  - Never make the leap to production
  - Unclear success criteria

Why do so many projects fail?

- ML is still research - you shouldn’t aim for 100% success rate
- But, many are doomed to fail:
  - Technically infeasible or poorly scoped
  - Never make the leap to production
  - Unclear success criteria
  - Poor team management

Module overview

Lifecycle

- How to think about all of the activities in an ML project

Prioritizing projects

- Assessing the feasibility and impact of your projects

Archetypes

- The main categories of ML projects, and the implications for project management

Metrics

- How to pick a single number to optimize

Baselines

- How to know if your model is performing well

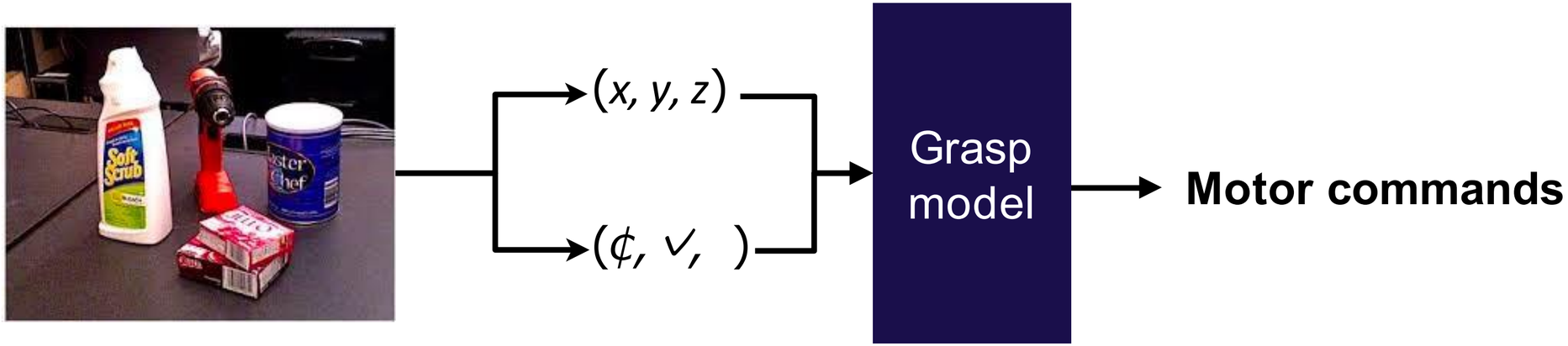
Running case study - pose estimation

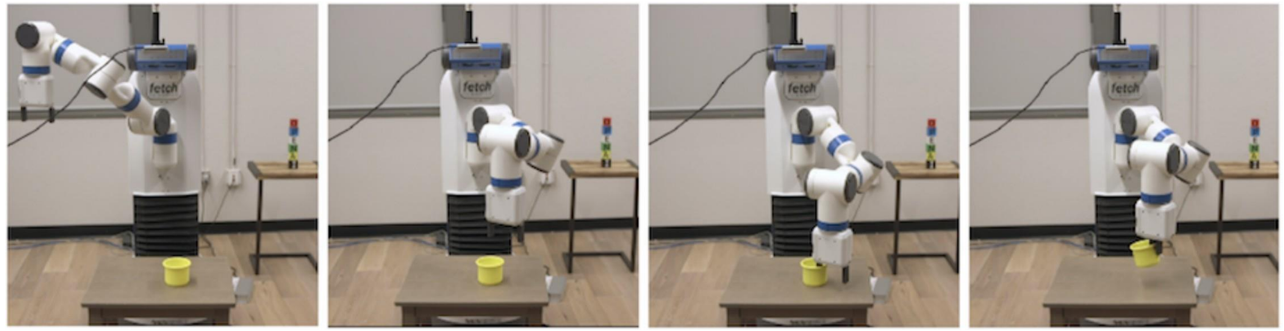
(x, y, z) **Position (L2 loss)**

$(¢, \checkmark, \ )$ **Orientation (L2 loss)**

Xiang, Yu, et al. "PoseCNN: A Convolutional Neural Network for 6D Object Pose Estimation in Cluttered Scenes." arXiv preprint arXiv:1711.00199 (2017).

Full Stack Robotics works on grasping





Module overview

Lifecycle

- **How to think about all of the activities in an ML project**
- **Assessing the feasibility and impact of your projects**
- **The main categories of ML projects, and the implications for project management**
- **How to pick a single number to optimize**
- **How to know if your model is performing well**

Lifecycle of a ML project

Planning & project setup

Lifecycle of a ML project

Planning & project setup

- Decide to work on pose estimation
- Determine requirements & goals
- Allocate resources
- Consider the ethical implications
- Etc.

Lifecycle of a ML project

Lifecycle of a ML project

Planning & project setup

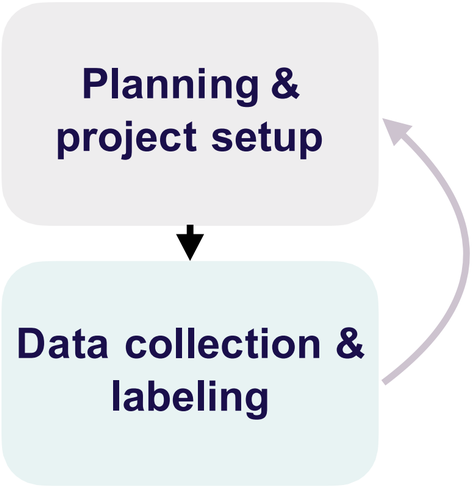
↓

Data collection & labeling

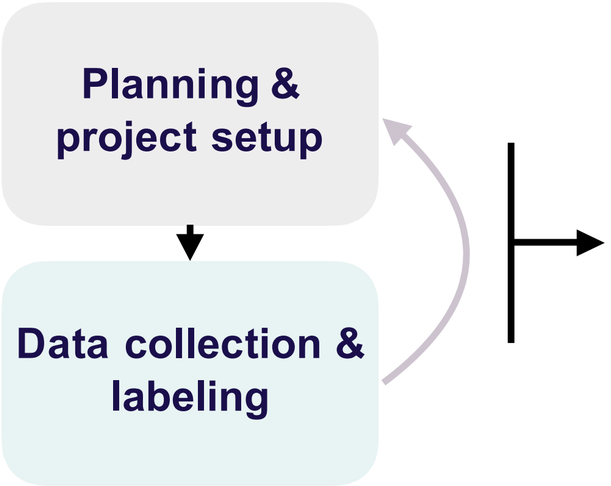
→

- Collect training objects
- Set up sensors (e.g., cameras)
- Capture images of objects
- Annotate with ground truth (how?)

Lifecycle of a ML project

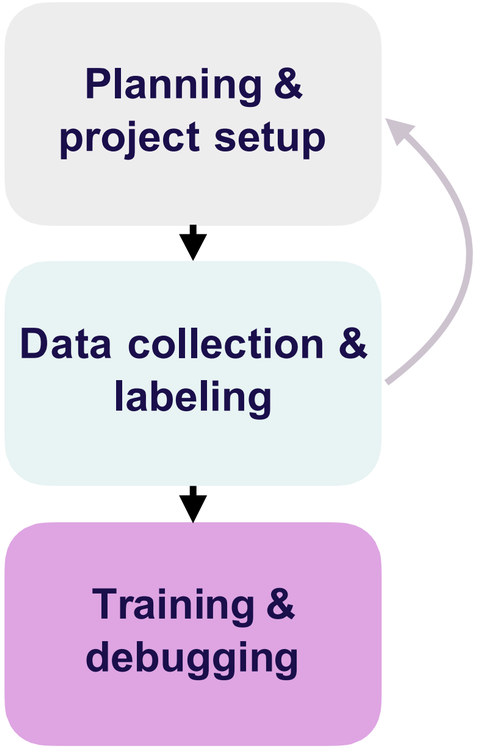


Lifecycle of a ML project

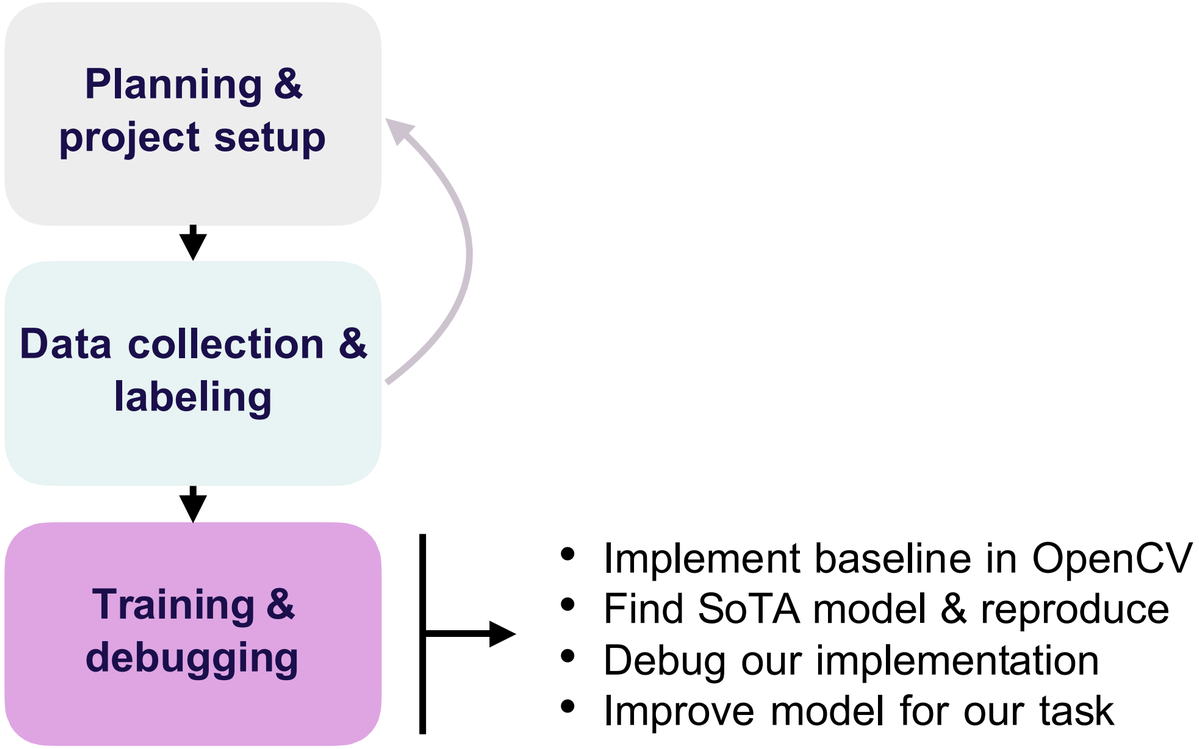


- Too hard to get data.
- Easier to label a different task (e.g., hard to annotate pose, easier to annotate per-pixel segmentation)

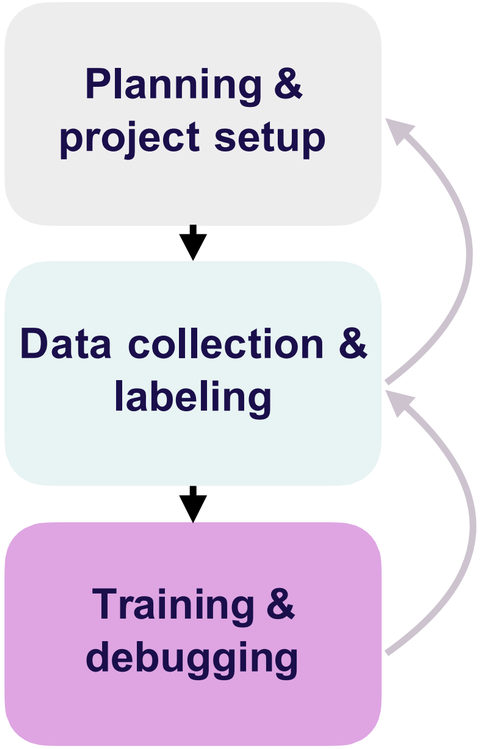
Lifecycle of a ML project



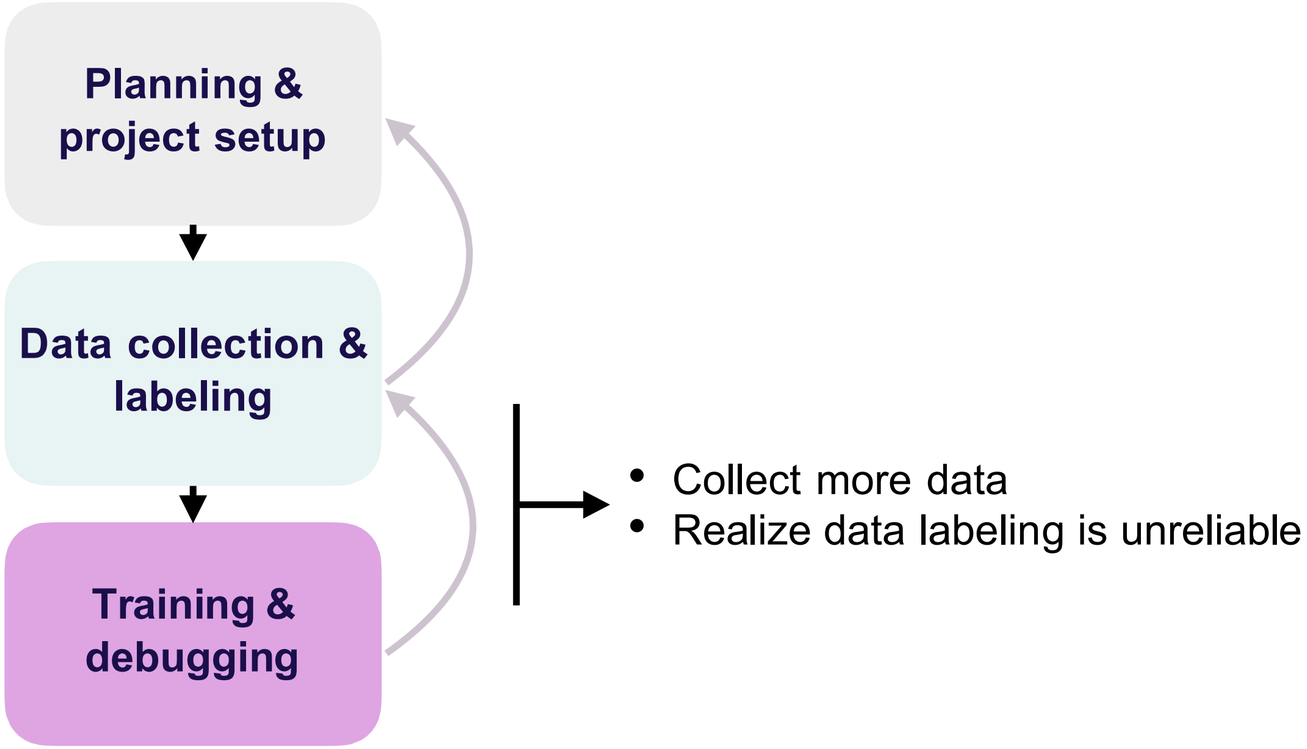
Lifecycle of a ML project



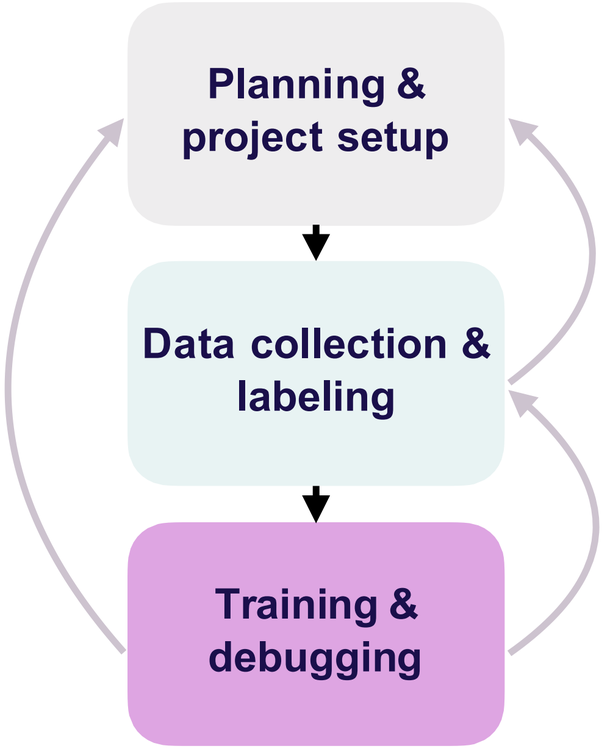
Lifecycle of a ML project



Lifecycle of a ML project

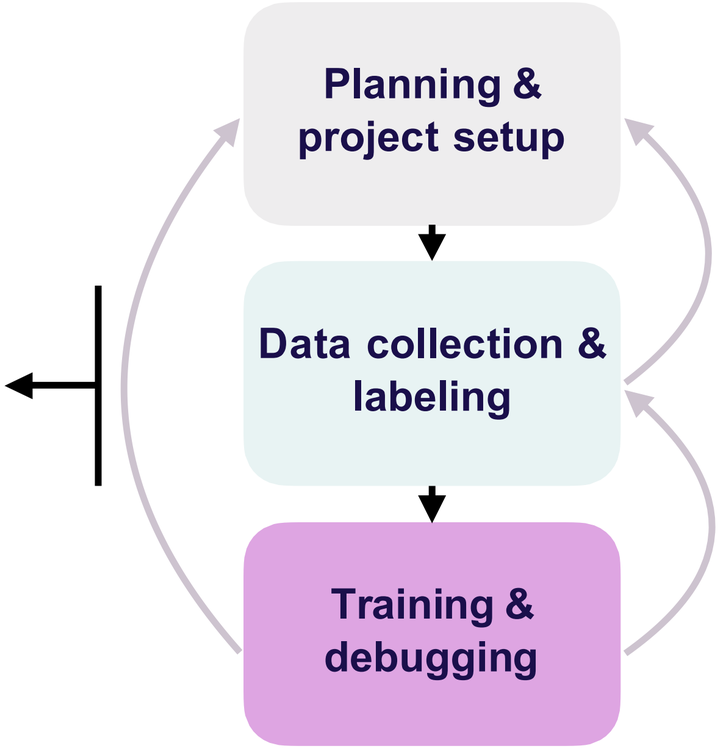


Lifecycle of a ML project

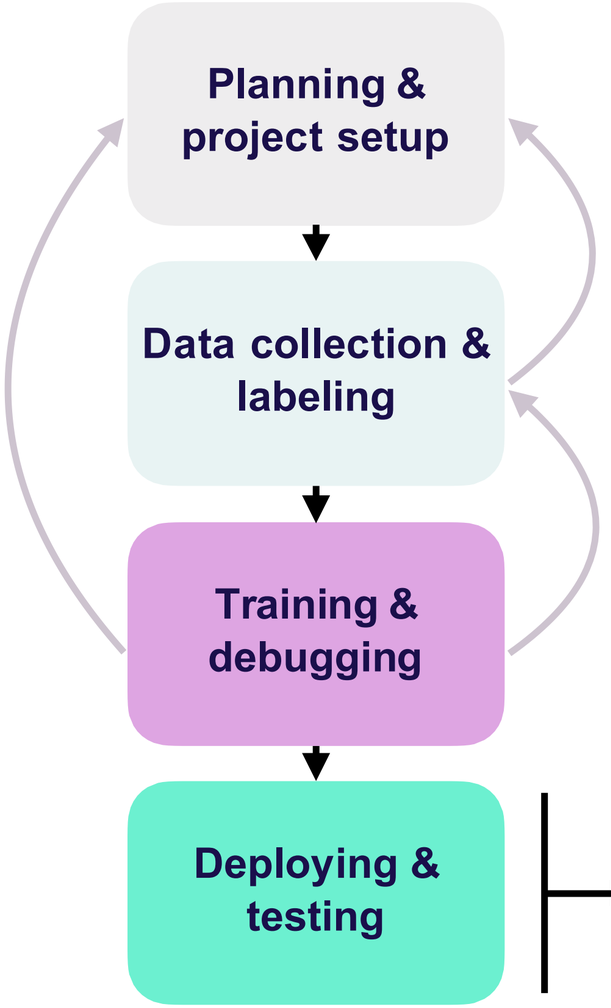


Lifecycle of a ML project

- Realize task is too hard
- Requirements trade off with each other - revisit which are most important

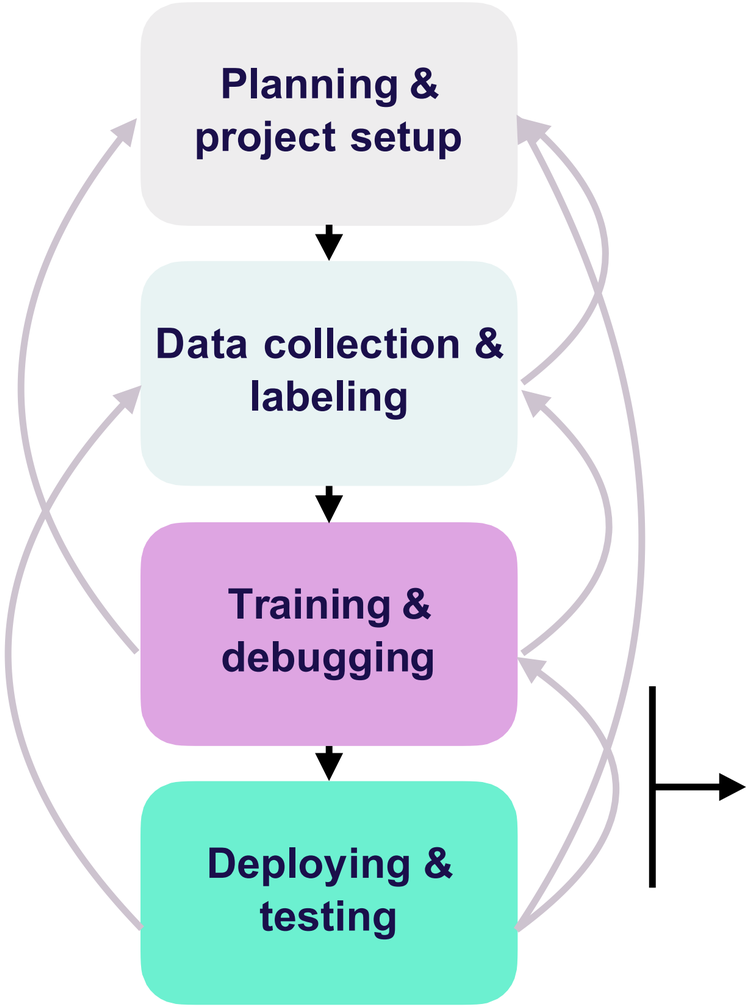


Lifecycle of a ML project



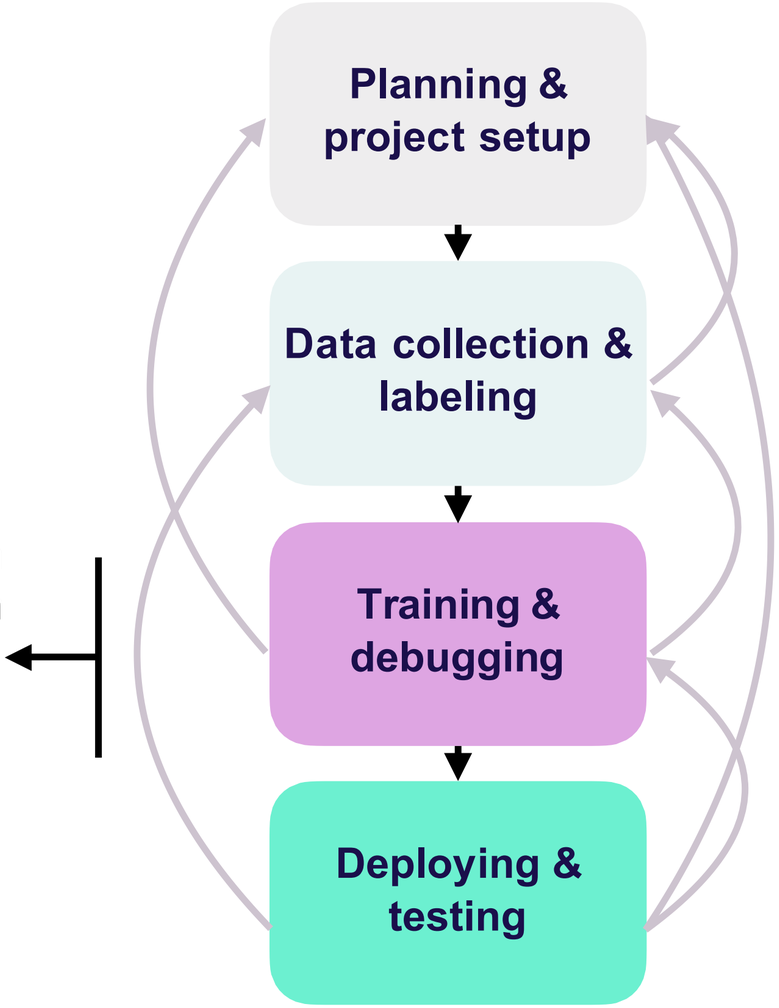
- Pilot in grasping system in the lab
- Write tests to prevent regressions and evaluate for biases
- Roll out in production

Lifecycle of a ML project



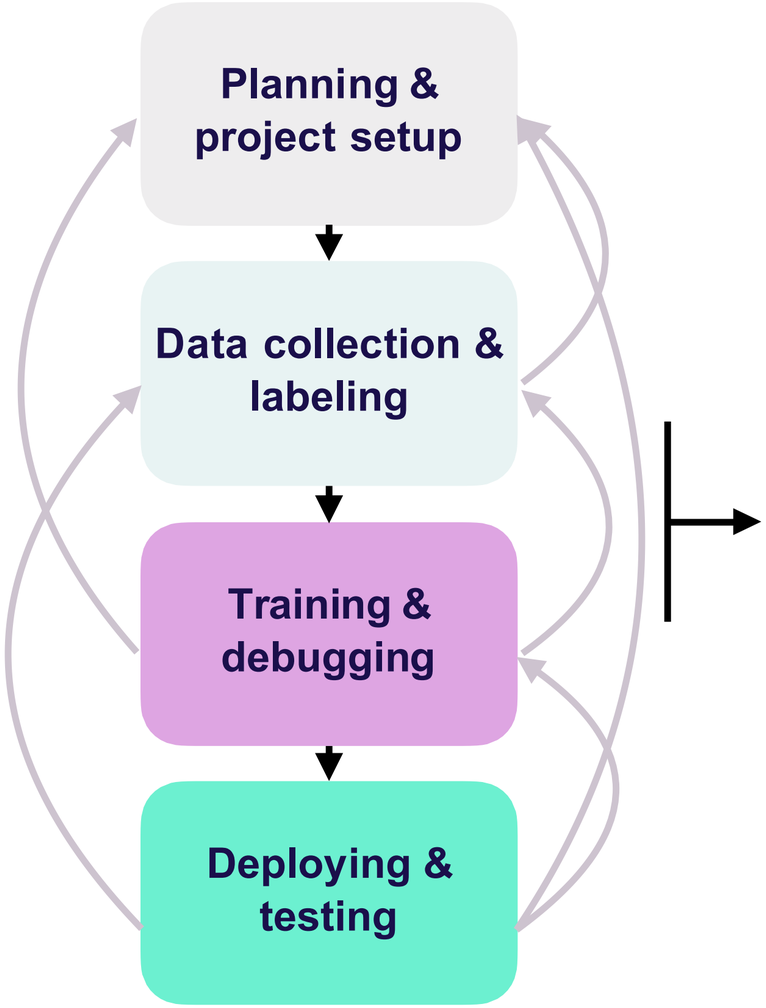
- Doesn't work in the lab - keep improving accuracy of model

Lifecycle of a ML project



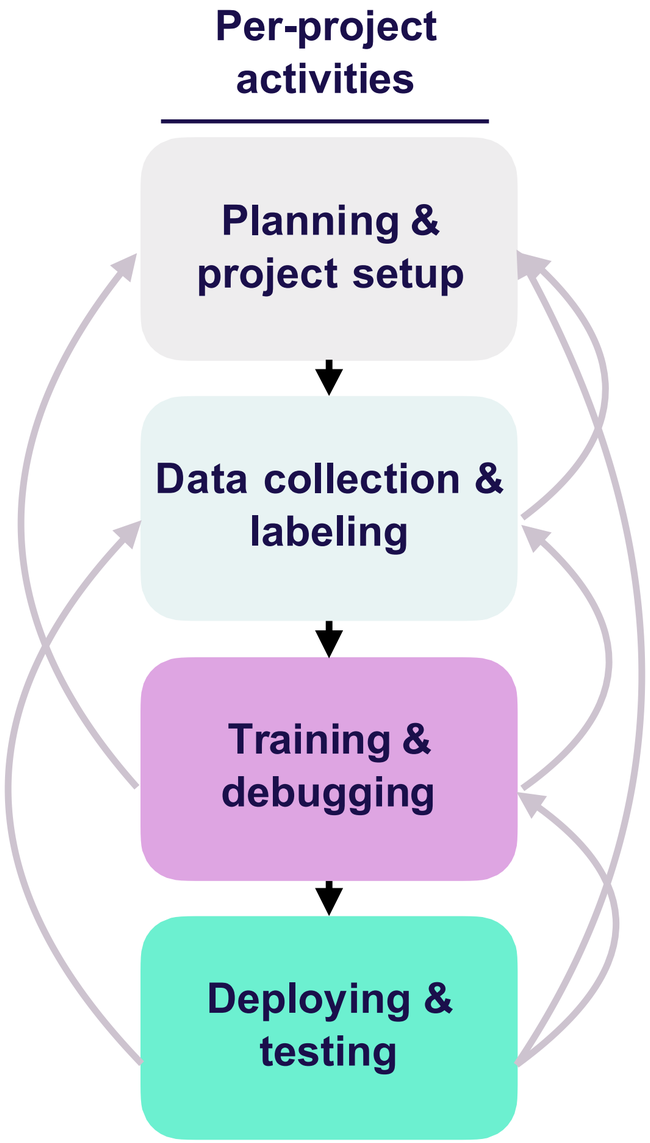
- Fix data mismatch between training data and data seen in deployment
- Collect more data
- Mine hard cases

Lifecycle of a ML project

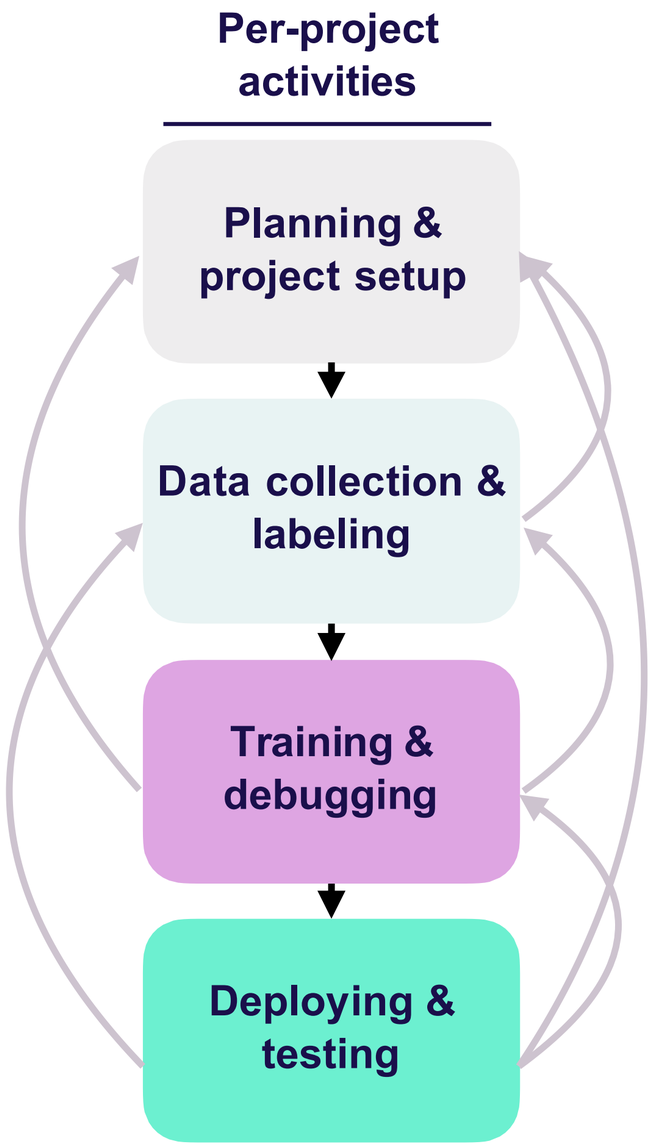


- The metric you picked doesn't actually drive downstream user behavior. Revisit the metric.
- Performance in the real world isn't great - revisit requirements (e.g., do we need to be faster or more accurate?

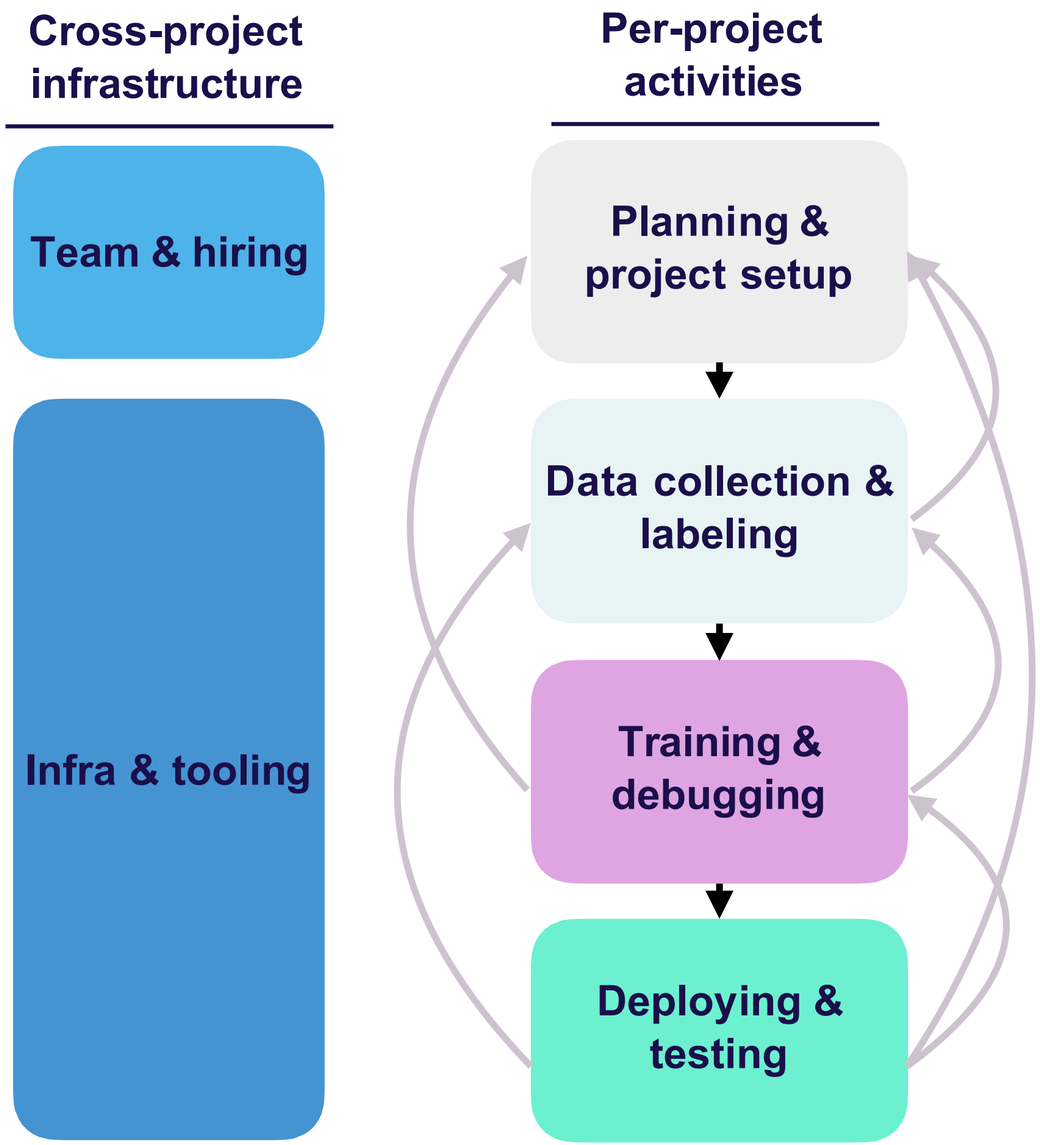
Lifecycle of a ML project



Lifecycle of a ML project



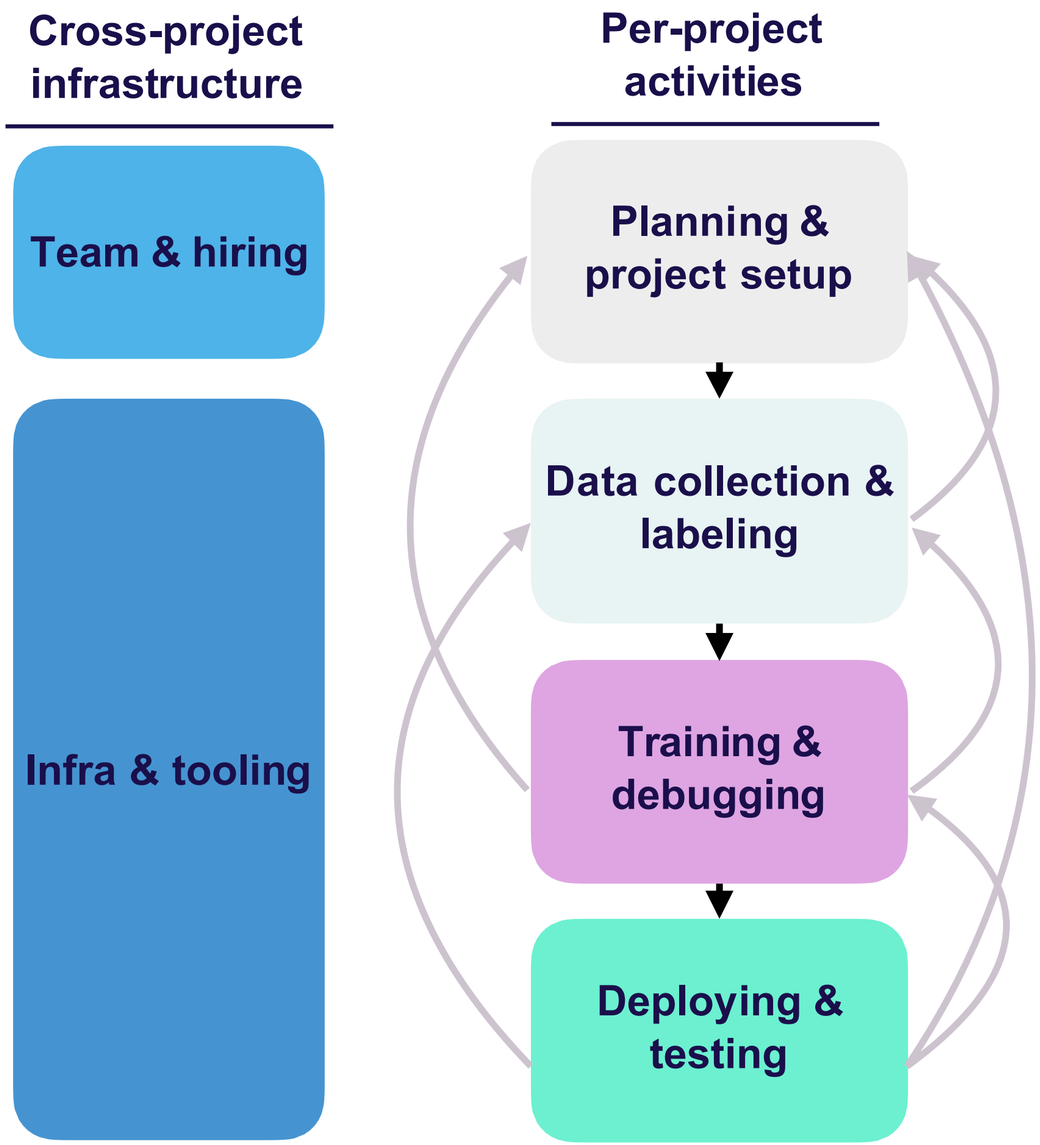
Lifecycle of a ML project



What else do you need to know?

- Understand state of the art in your domain
  - Understand what's possible
  - Know what to try next
- We will introduce most promising research areas

Lifecycle of a ML project



Questions?

Module overview

- How to think about all of the activities in an ML project

Prioritizing projects

- **Assessing the feasibility and impact of your projects**

- The main categories of ML projects, and the implications for project management

Metrics

- How to pick a single number to optimize

- How to know if your model is performing well

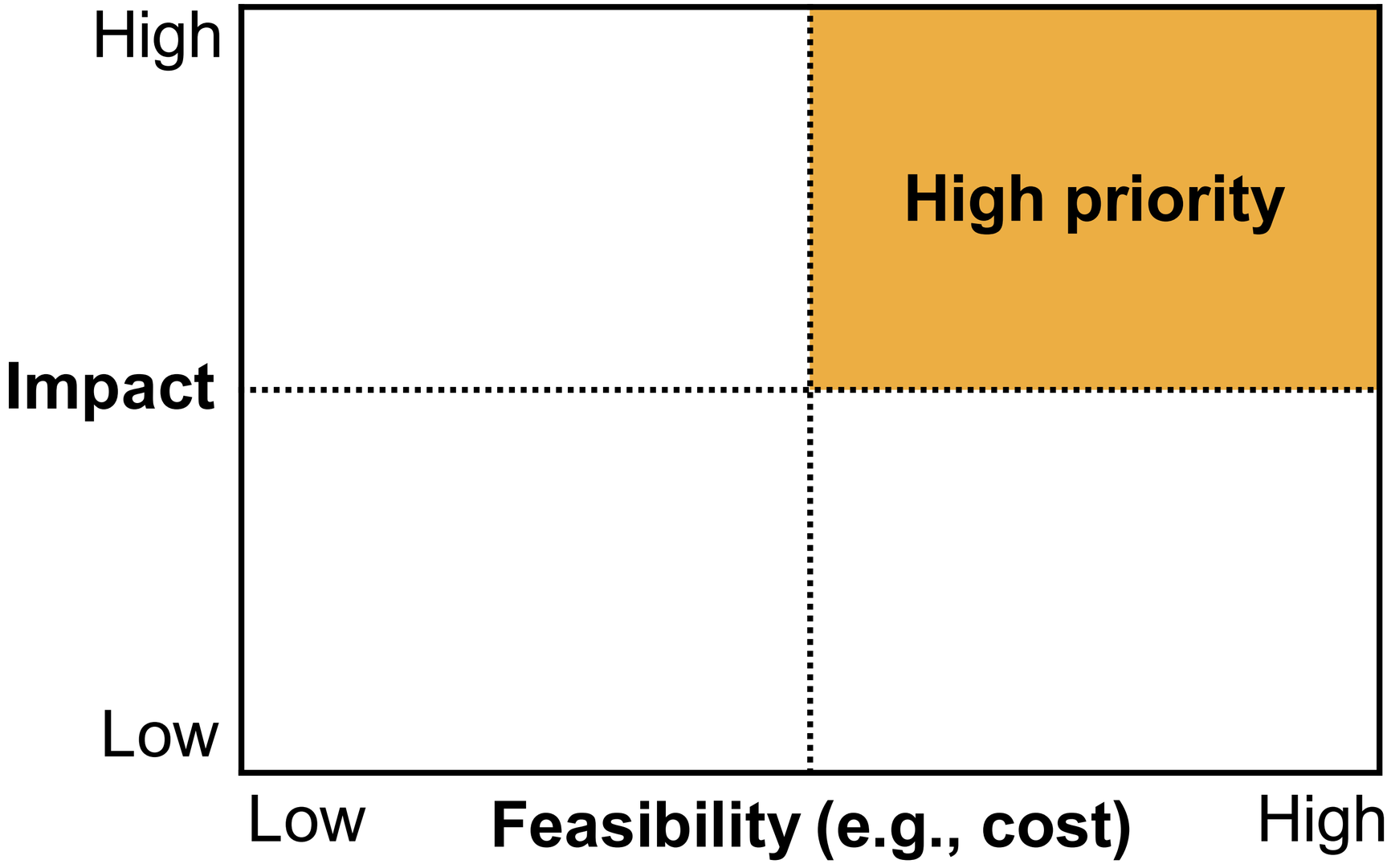
Key points for prioritizing projects

A. High-impact ML problems

- Friction in your product
- Complex parts of your pipeline
- Places where cheap prediction is valuable
- What else are people doing?

B. Cost of ML projects is driven by data availability. Also consider accuracy requirements and intrinsic difficulty of the problem

A (general) framework for prioritizing projects



Mental models for high-impact ML projects

1. Where can you take advantage of cheap prediction?
2. Where is there friction in your product?
3. Where can you automate complicated manual processes?
4. What are other people doing?

What does ML make economically feasible?

The economics of AI
(Agrawal, Gans, Goldfarb)

- AI reduces cost of prediction
- Prediction is central for decision making
- Cheap prediction means
  - Prediction will be everywhere
  - Even in problems where it was too expensive before (e.g., for most people, hiring a driver)
- **Implication:** Look for projects where cheap prediction will have a huge business impact

Prediction Machines: The Simple Economics of Artificial Intelligence (Agrawal, Gans, Goldfarb)

What does your product need?

"Discover Weekly removed the friction of chasing everything down yourself and instead brought the music to you in a neat little package every Monday morning."

NOTED

Three Principles for Designing ML-Powered Products

October 2019

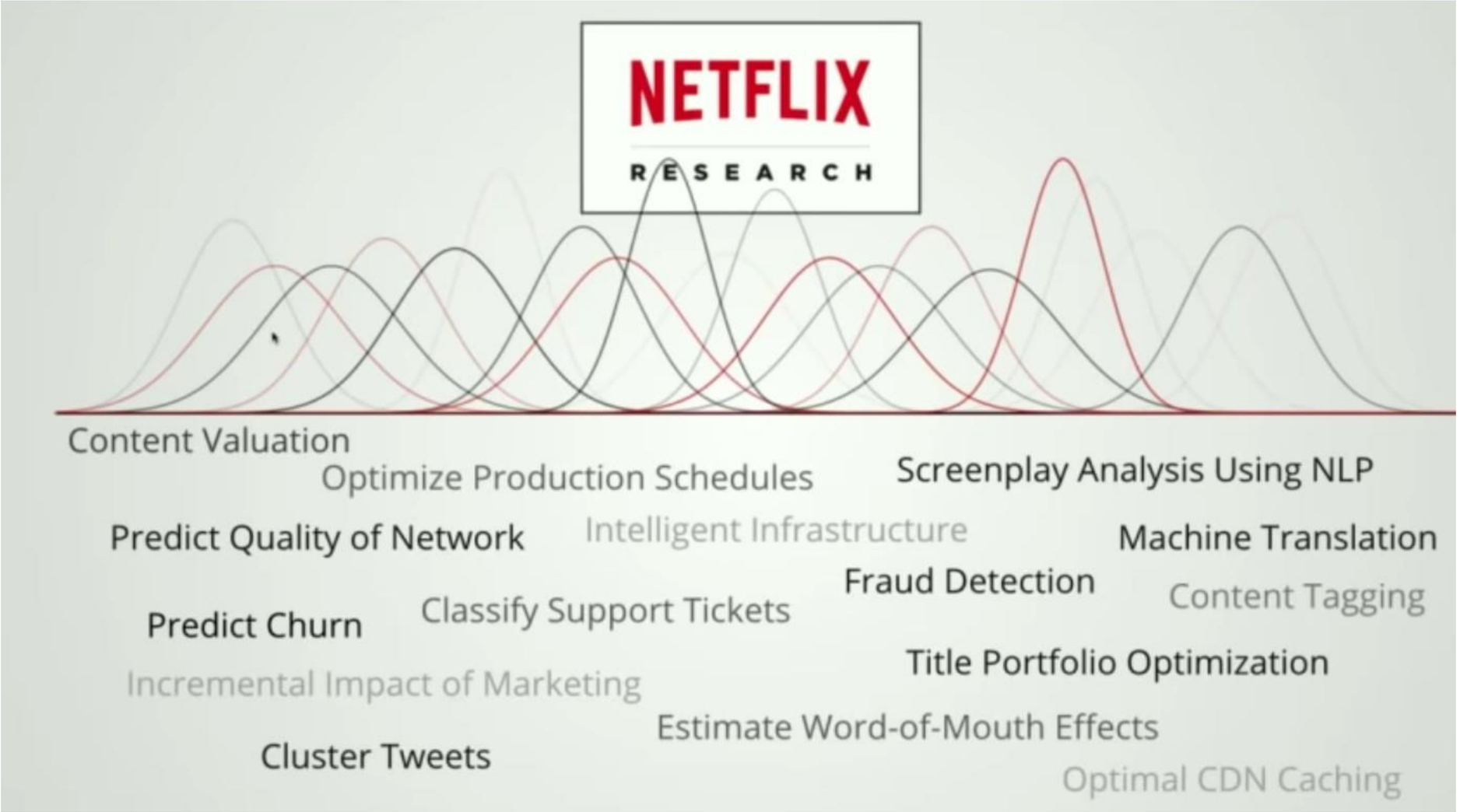
What is ML good at?

Software 2.0
(Andrej Karpathy)

- *Software 1.0* = traditional programs with explicit instructions (python / c++ / etc)
- Software 2.0 = humans specify goals, and algorithm searches for a program that works
- 2.0 programmers work with datasets, which get compiled via optimization
- Why? Works better, more general, computational advantages
- **Implication:** look for complicated rule-based software where we can learn the rules instead of programming them

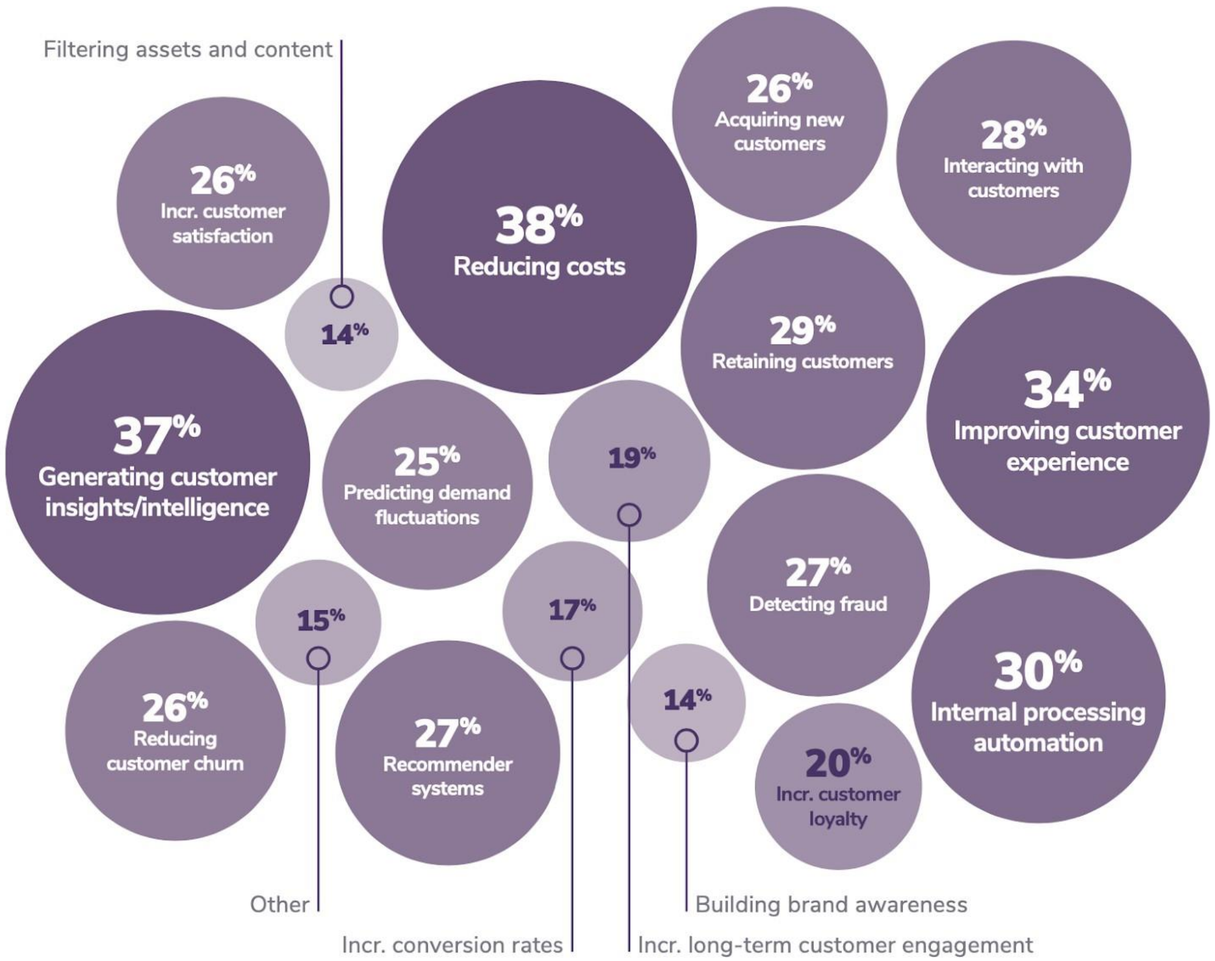
Software 2.0 (Andrej Karpathy): https://medium.com/@karpathy/software-2-0-a64152b37c35

What are other people doing?



Human-Centric Machine Learning Infrastructure @Netflix (Ville Tuulos, InfoQ 2019)

What are other people doing?



2020 state of enterprise machine learning (Algorithmia, 2020)

What are other people doing?

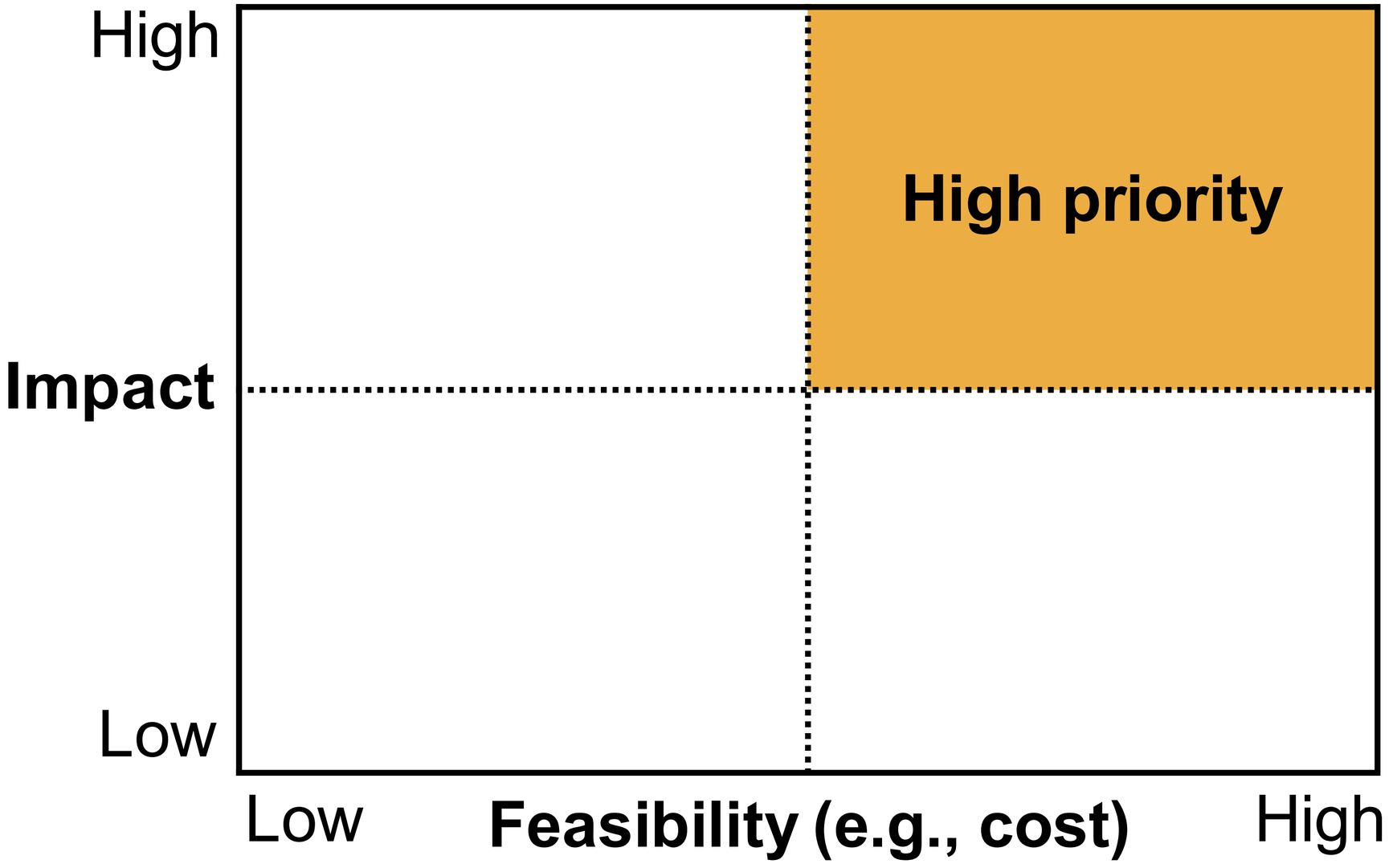
- Papers from Google, Facebook, Nvidia, Netflix, etc
- Blog posts from top earlier-stage companies (Uber, Lyft, Spotify, Stripe, etc)

Case studies

- Using Machine Learning to Predict Value of Homes On Airbnb (Robert Chang, Airbnb Engineering & Data Science, 2017)
- Using Machine Learning to Improve Streaming Quality at Netflix (Chaitanya Ekanadham, Netflix Technology Blog, 2018)
- 150 Successful Machine Learning Models: 6 Lessons Learned at Booking.com (Bernardi et al., KDD, 2019)Asdf
- How we grew from 0 to 4 million women on our fashion app, with a vertical machine learning approach (Gabriel Aldamiz, HackerNoon, 2018)
- Machine Learning-Powered Search Ranking of Airbnb Experiences (Mihajlo Grbovic, Airbnb Engineering & Data Science, 2019)
- From shallow to deep learning in fraud (Hao Yi Ong, Lyft Engineering, 2018)
- Space, Time and Groceries (Jeremy Stanley, Tech at Instacart, 2017)
- Creating a Modern OCR Pipeline Using Computer Vision and Deep Learning (Brad Neuberg, Dropbox Engineering, 2017)
- Scaling Machine Learning at Uber with Michelangelo (Jeremy Hermann and Mike Del Balso, Uber Engineering, 2019)
- Spotify's Discover Weekly: How machine learning finds your new music (Umesh .A Bhat, 2017)

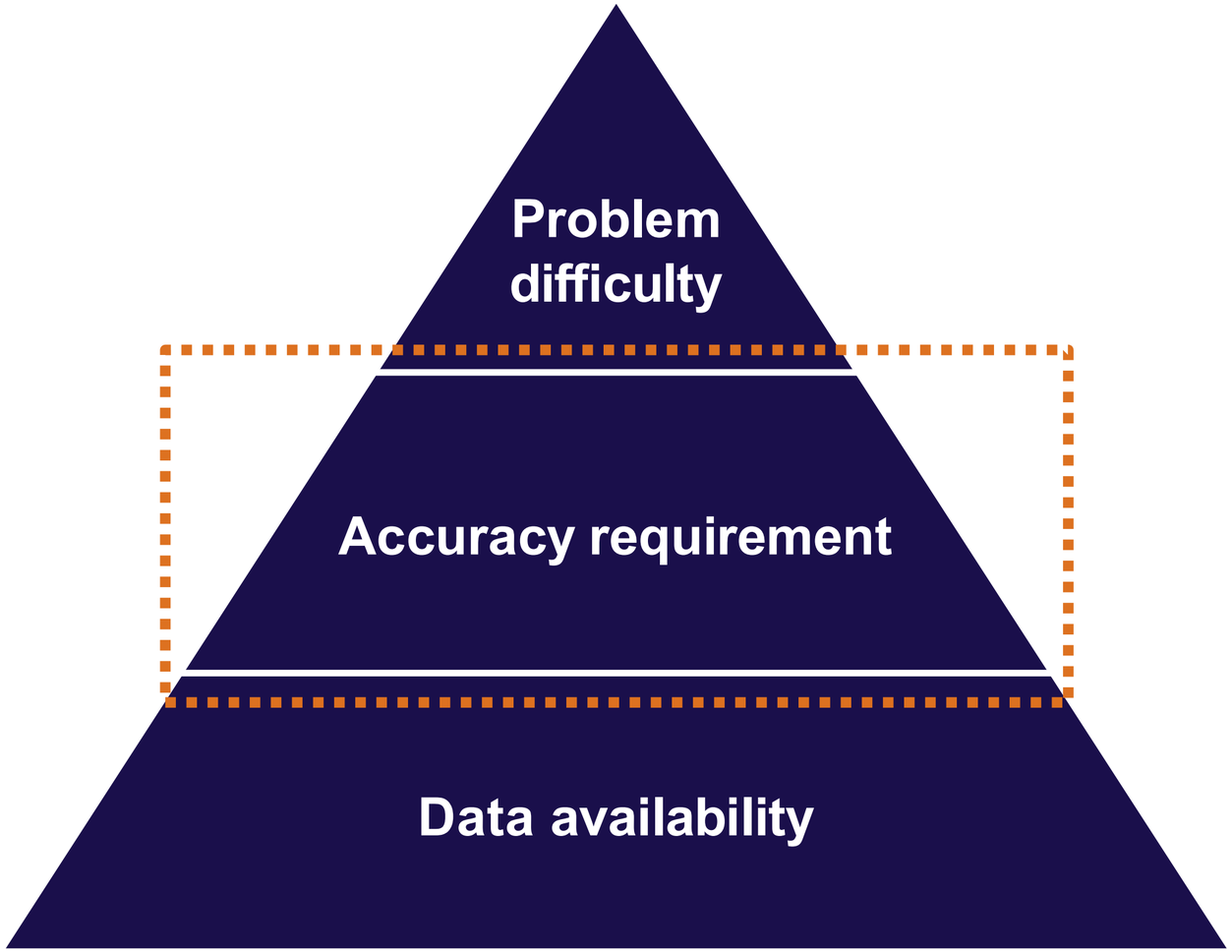
Credit for compiling this list: Chip Huyen (Machine Learning Systems Design Lecture 2 note)

A (general) framework for prioritizing projects



Assessing feasibility of ML projects

Cost drivers



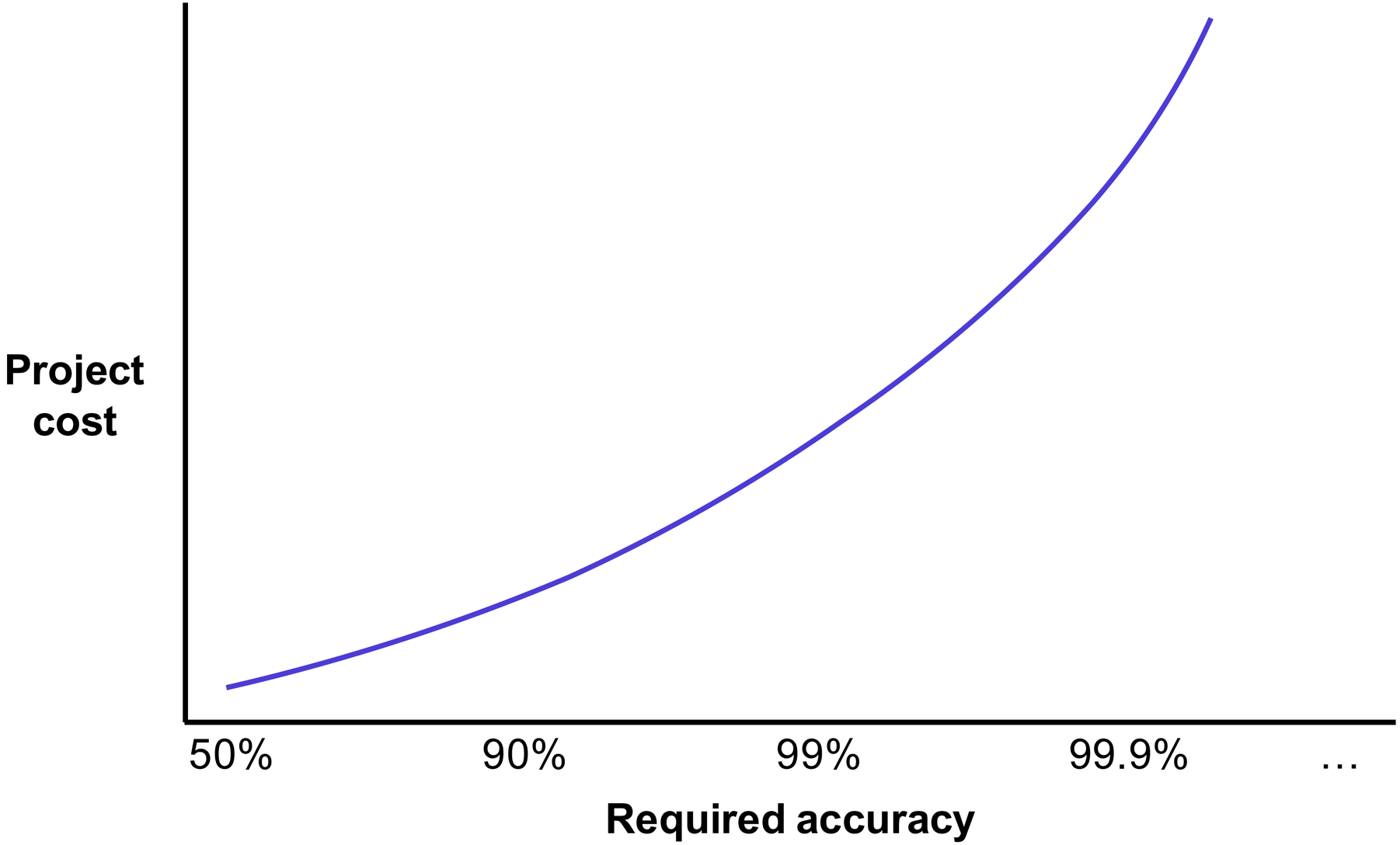
Main considerations

- Is the problem well-defined?
- Good published work on similar problems?
  (newer problems mean more risk & more technical effort)
- Compute requirements?
- Can a human do it?

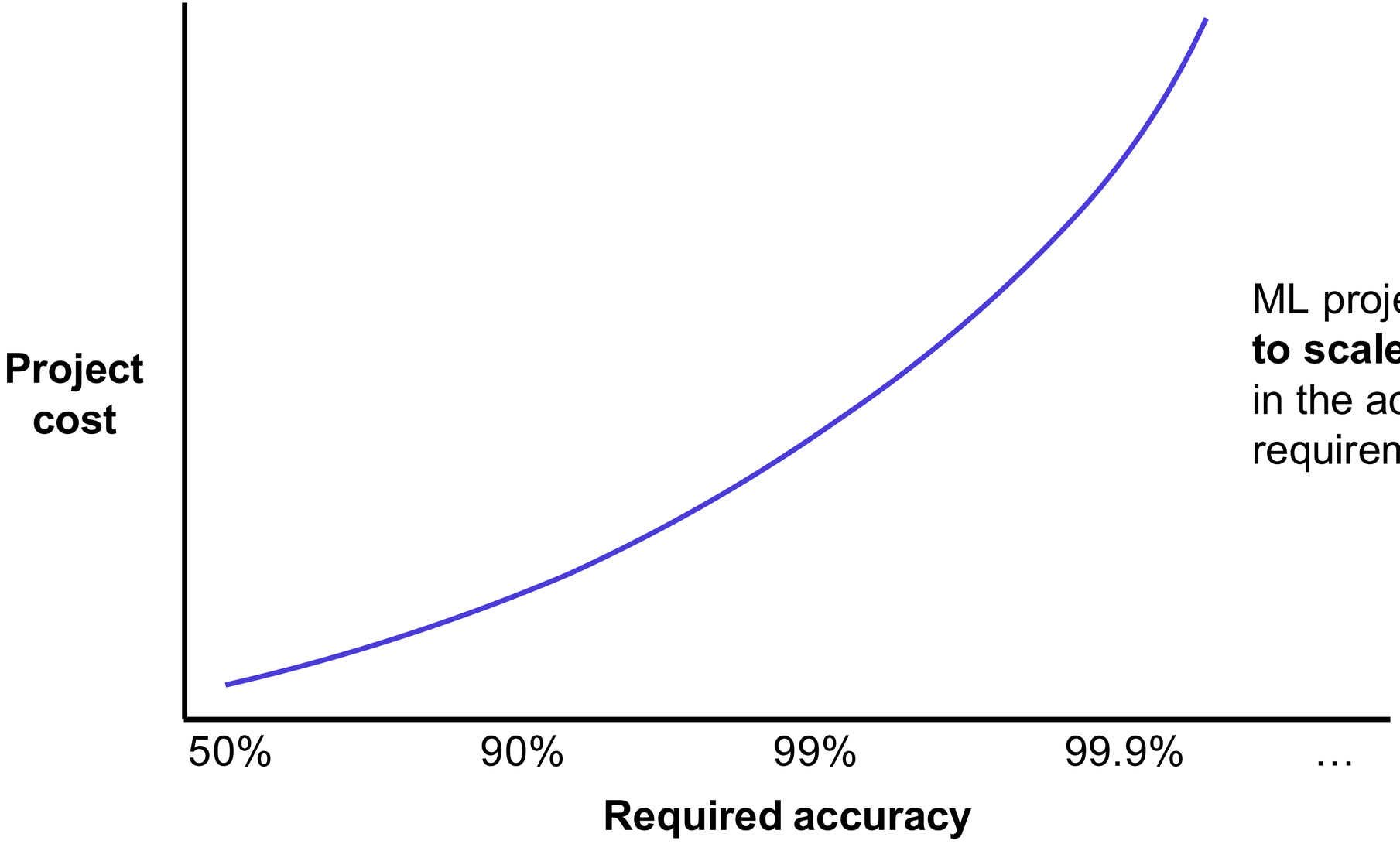
- How costly are wrong predictions?
- How frequently does the system need to be right to be useful?
- Ethical implications?

- How hard is it to acquire data?
- How expensive is data labeling?
- How much data will be needed?
- How stable is the data?
- Data security requirements?

Why are accuracy requirements so important?



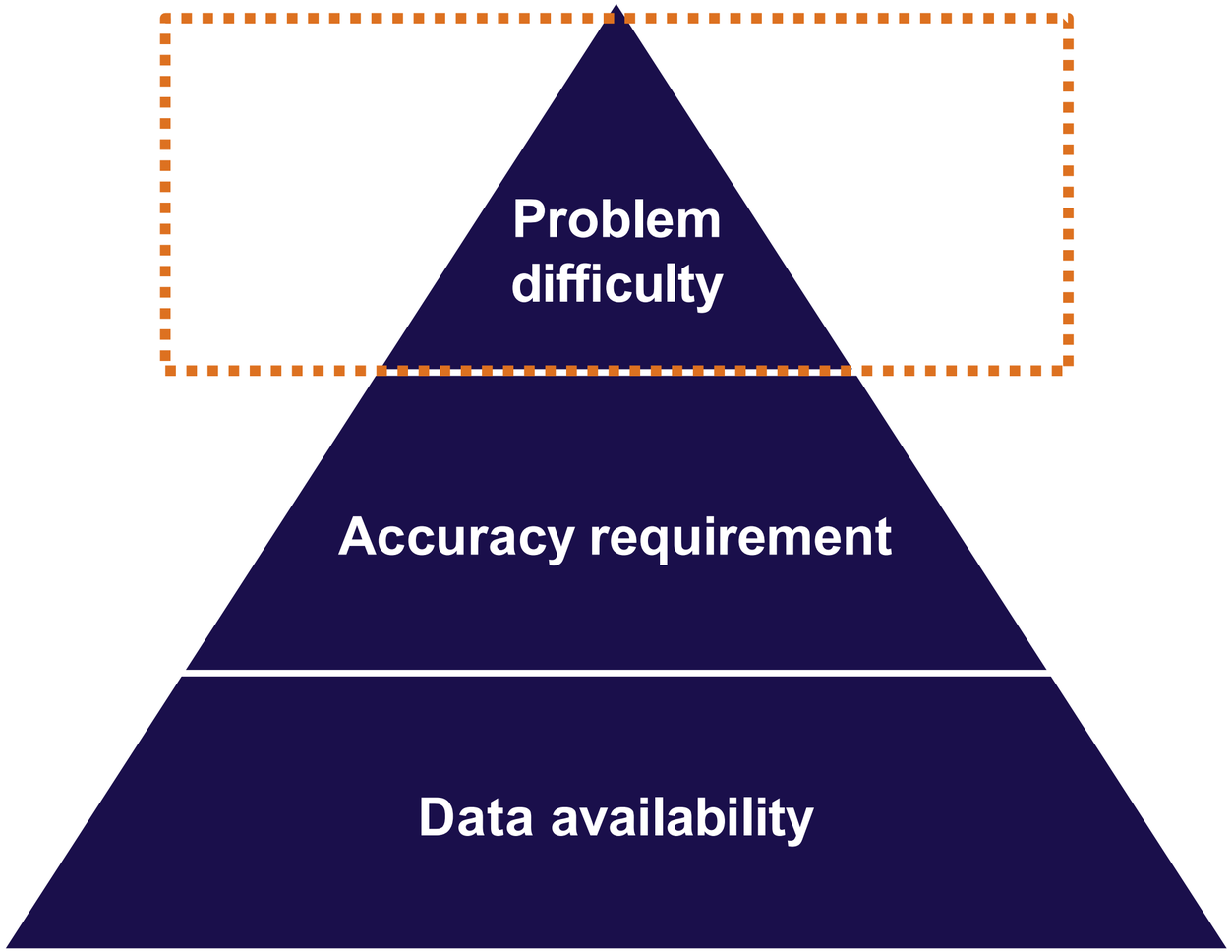
Why are accuracy requirements so important?



ML project **costs tend to scale super-linearly** in the accuracy requirement

Assessing feasibility of ML projects

Cost drivers



Main considerations

- Is the problem well-defined?
- Good published work on similar problems?
  (newer problems mean more risk & more technical effort)
- Compute requirements?
- Can a human do it?

- How costly are wrong predictions?
- How frequently does the system need to be right to be useful?
- Ethical implications?

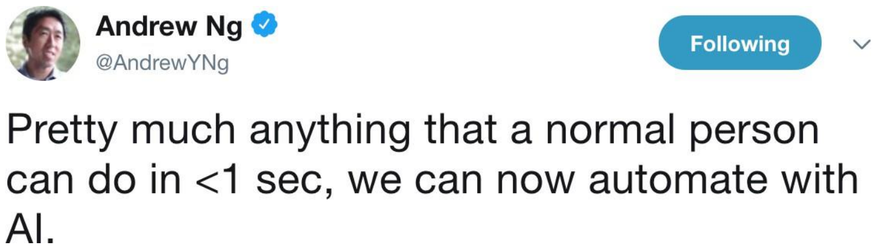
- How hard is it to acquire data?
- How expensive is data labeling?
- How much data will be needed?
- How stable is the data?
- Data security requirements?

What’s still hard in machine learning?

"It may be a hundred years before a computer beats humans at Go -- maybe even longer," said Dr. Piet Hut, an astrophysicist at the Institute for Advanced Study in Princeton, N.J., and a fan of the game. "If a reasonably intelligent person learned to play Go, in a few months he could beat all existing computer programs. You don't have to be a Kasparov."

New York Times, July 1997

What’s still hard in machine learning?



Examples	Counter-examples?
• Recognize content of images • Understand speech • Translate speech • Grasp objects • etc.	• Understand humor / sarcasm • In-hand robotic manipulation • Generalize to new scenarios • etc.

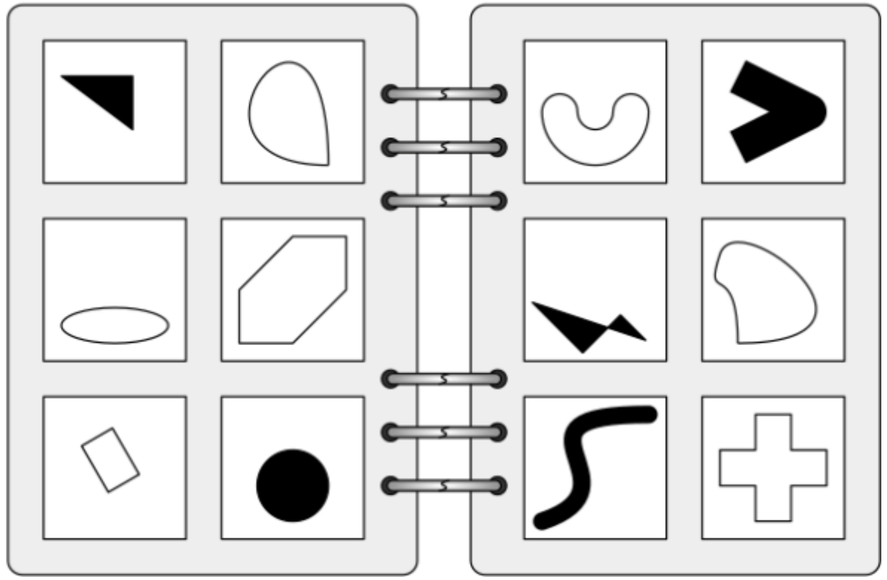
What’s still hard in machine learning?

- Unsupervised learning
- Reinforcement learning
- Both are showing promise in limited domains where tons of data and compute are available

What’s still hard in supervised learning?

- Answering questions
- Summarizing text
- Predicting video
- Building 3D models
- Real-world speech recognition
- Resisting adversarial examples
- Doing math
- Solving word puzzles
- Bongard problems
- Etc

Example of a Bongard Problem



What types of problems are hard?

	Instances	Examples
Output is complex	• High-dimensional output • Ambiguous output	• 3D reconstruction • Video prediction • Dialog systems • Open-ended recommender systems
Reliability is required	• High precision is required • Robustness is required	• Failing safely out-of-distribution • Robustness to adversarial attacks • High-precision pose estimation
Generalization is required	• Out of distribution data • Reasoning, planning, causality	• Self-driving: edge cases • Self-driving: control • Small data

Why is FSR focusing on pose estimation?

Impact

- FSR's goal is grasping - requires reliable pose estimation
- Traditional robotics pipeline uses hand-designed heuristics & online optimization
  - Slow
  - Brittle
  - Great candidate for Software 2.0!

Feasibility

- Data availability
  - Easy to collect data
  - Labeling data could be a challenge, but can instrument lab with sensors
- Accuracy requirement
  - Require high accuracy to grasp an object: <0.5cm
  - However, low cost of failure - picks per hour important, not % successes
- Problem difficulty
  - Similar published results exist but need to adapt to our objects and robot

How to run a ML feasibility assessment

A. Are you sure you need ML at all?

B. Put in the work up-front to define success criteria with all of the stakeholders

C. Consider the ethics of using ML

D. Do a literature review

E. Try to rapidly build a labeled benchmark dataset

F. Build a *minimal* viable product (e.g., manual rules)

G. Are you sure you need ML at all?

Questions?

Module overview

- How to think about all of the activities in an ML project
- Assessing the feasibility and impact of your projects

Archetypes

- **The main categories of ML projects, and the implications for project management**
- How to pick a single number to optimize
- How to know if your model is performing well

Machine learning product archetypes

	Examples
Software 2.0	• Improve code completion in an IDE • Build a customized recommendation system • Build a better video game AI

Machine learning product archetypes

	Examples
Software 2.0	• Improve code completion in an IDE • Build a customized recommendation system • Build a better video game AI
Human-in-the-loop	• Turn sketches into slides • Email auto-completion • Help a radiologist do their job faster

Machine learning product archetypes

	Examples
Software 2.0	• Improve code completion in an IDE • Build a customized recommendation system • Build a better video game AI
Human-in-the-loop	• Turn sketches into slides • Email auto-completion • Help a radiologist do their job faster
Autonomous systems	• Full self-driving • Automated customer support • Automated website design

Machine learning product archetypes

Software 2.0

Key questions

- Do your models truly improve performance?
- Does performance improvement generate business value?
- Do performance improvements lead to a data flywheel?

Machine learning product archetypes

	Key questions
Software 2.0	• Do your models truly improve performance? • Does performance improvement generate business value? • Do performance improvements lead to a data flywheel?
Human-in-the-loop	• How good does the system need to be to be useful? • How can you collect enough data to make it that good?

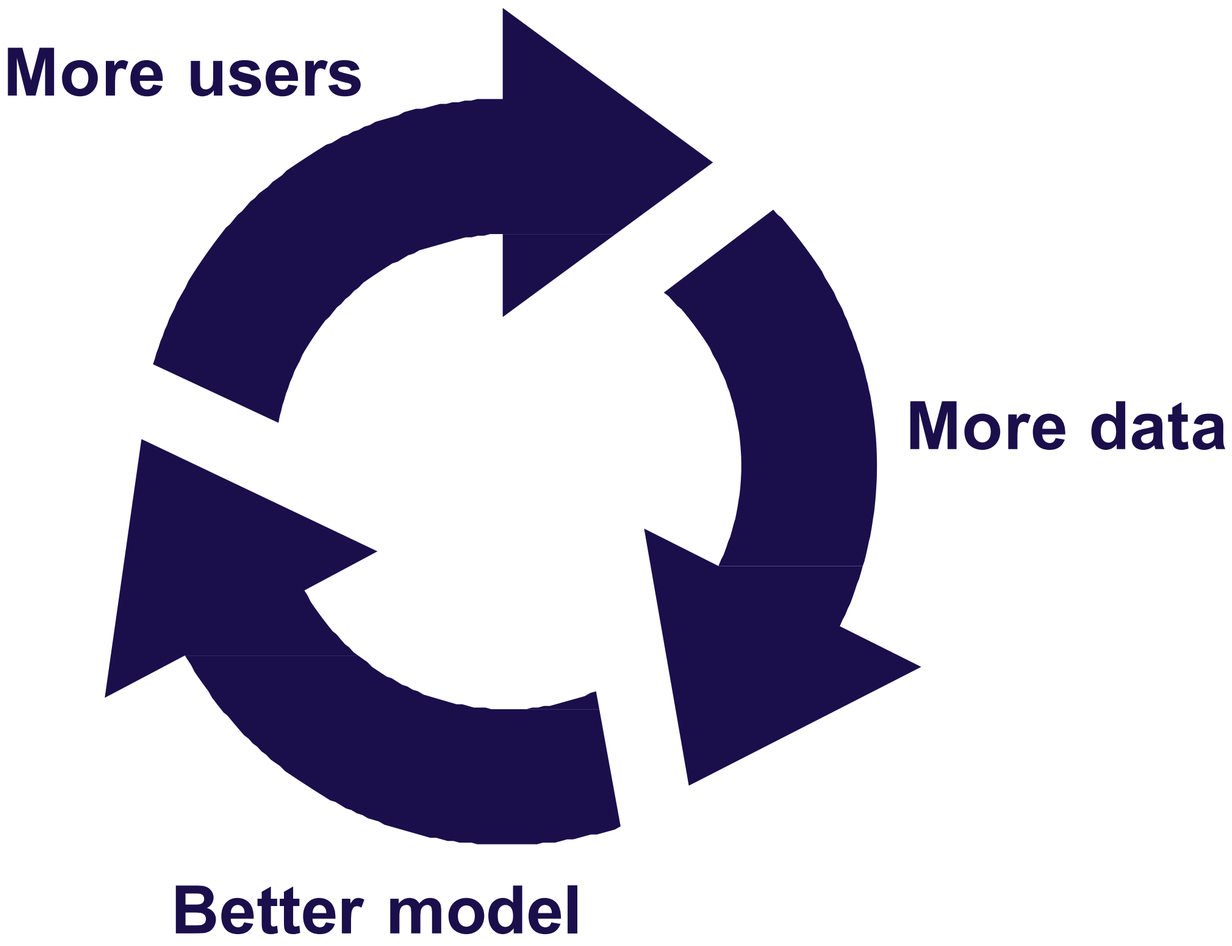
Machine learning product archetypes

	Key questions
Software 2.0	• Do your models truly improve performance? • Does performance improvement generate business value? • Do performance improvements lead to a data flywheel?
Human-in-the-loop	• How good does the system need to be to be useful? • How can you collect enough data to make it that good?
Autonomous systems	• What is an acceptable failure rate for the system? • How can you guarantee that it won’t exceed that failure rate? • How inexpensively can you label data from the system?

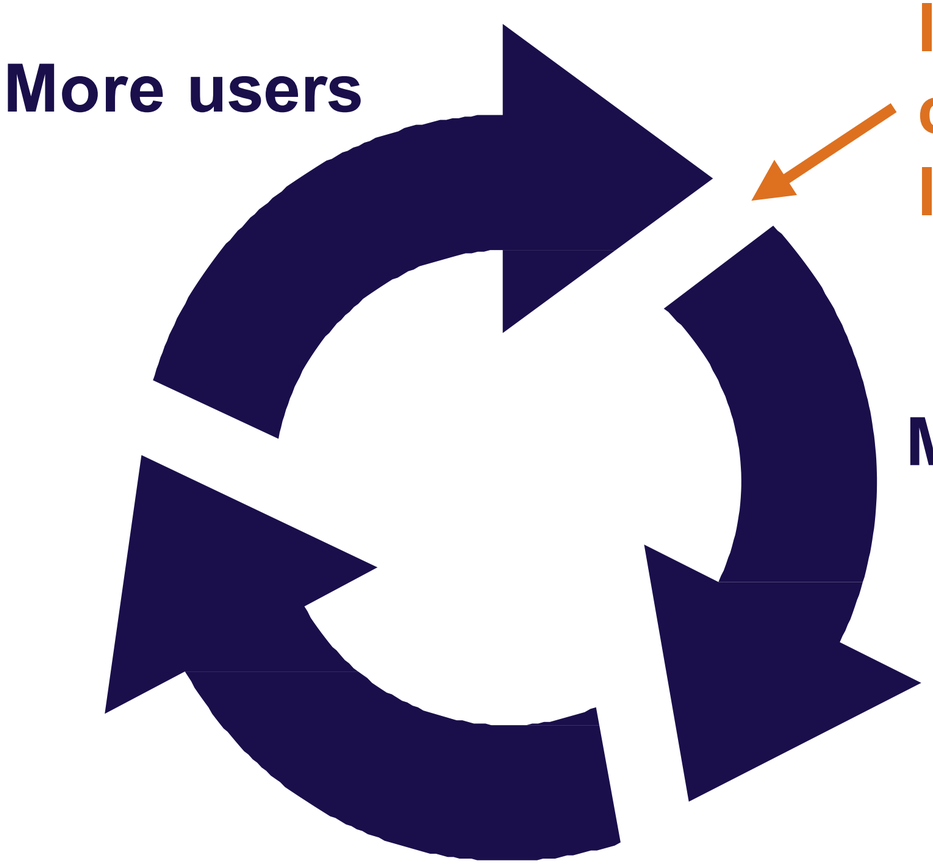
Machine learning product archetypes

	Key questions
Software 2.0	• Do your models truly improve performance? • Does performance improvement generate business value? • Do performance improvements lead to a data flywheel?
Human-in-the-loop	• How good does the system need to be to be useful? • How can you collect enough data to make it that good?
Autonomous systems	• What is an acceptable failure rate for the system? • How can you guarantee that it won't exceed that failure rate? • How inexpensively can you label data from the system?

Data flywheels

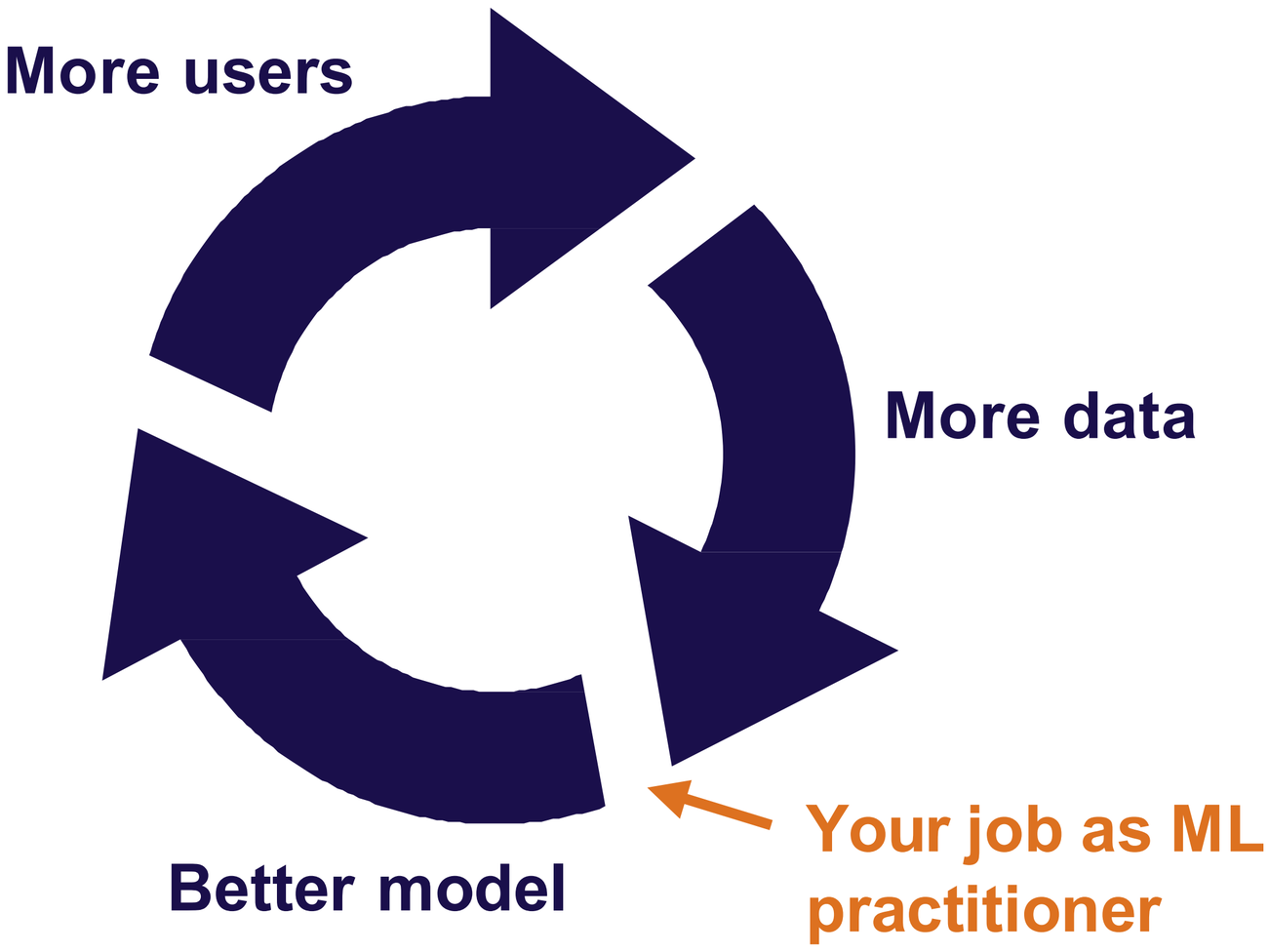


Data flywheels

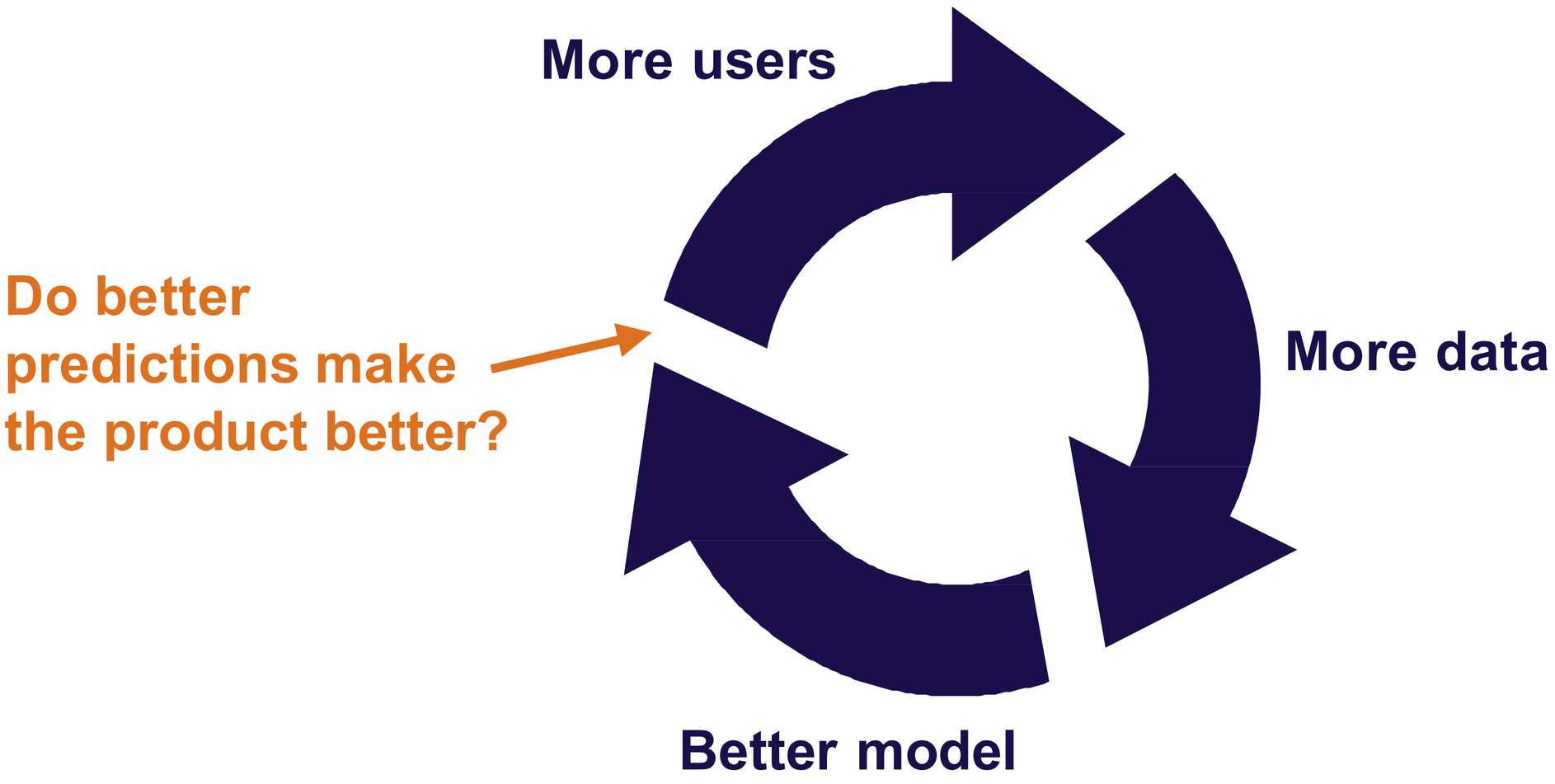


Do you have a data loop? Automatically collect data (and ideally labels) from your users

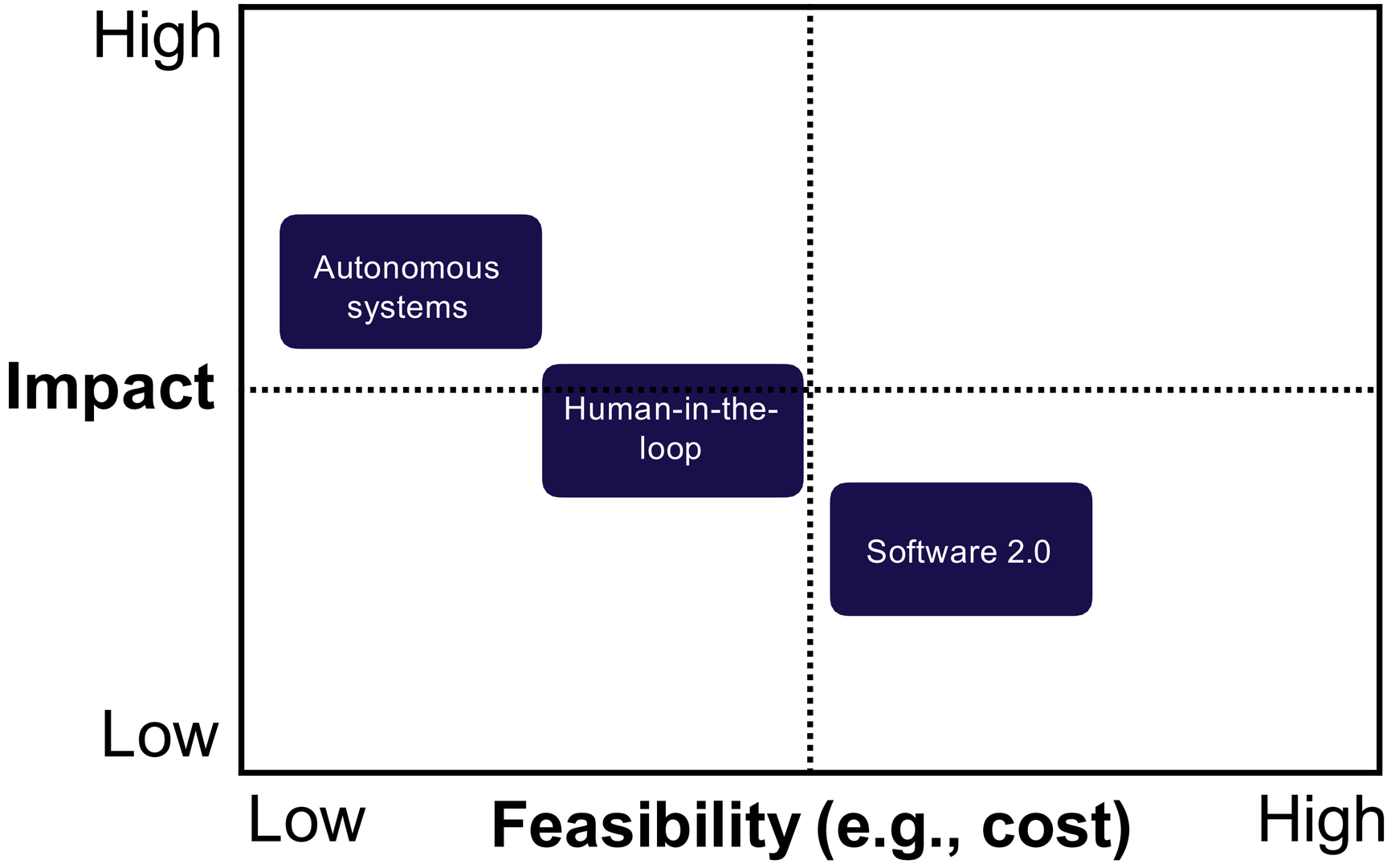
Data flywheels



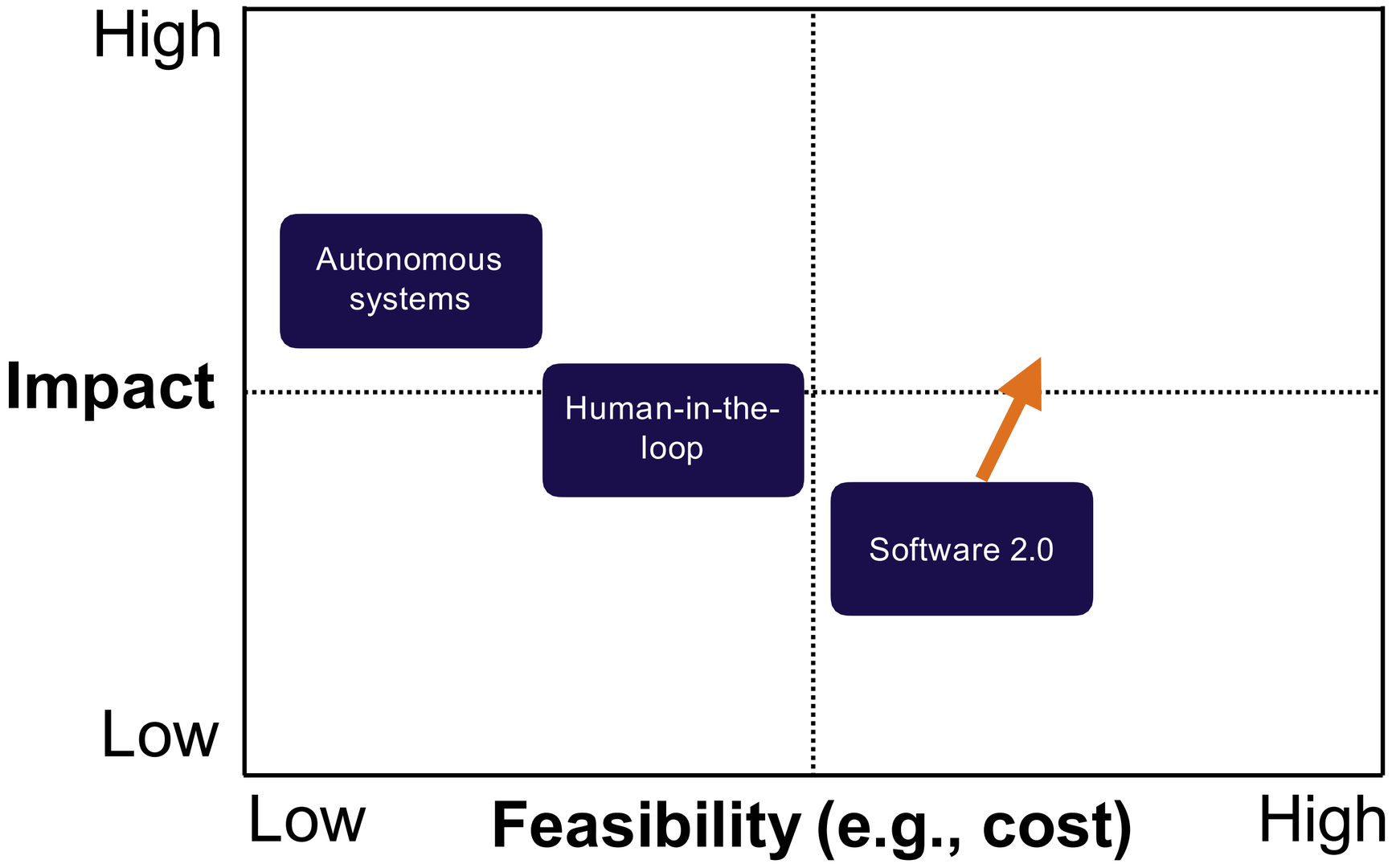
Data flywheels



Machine learning project archetypes

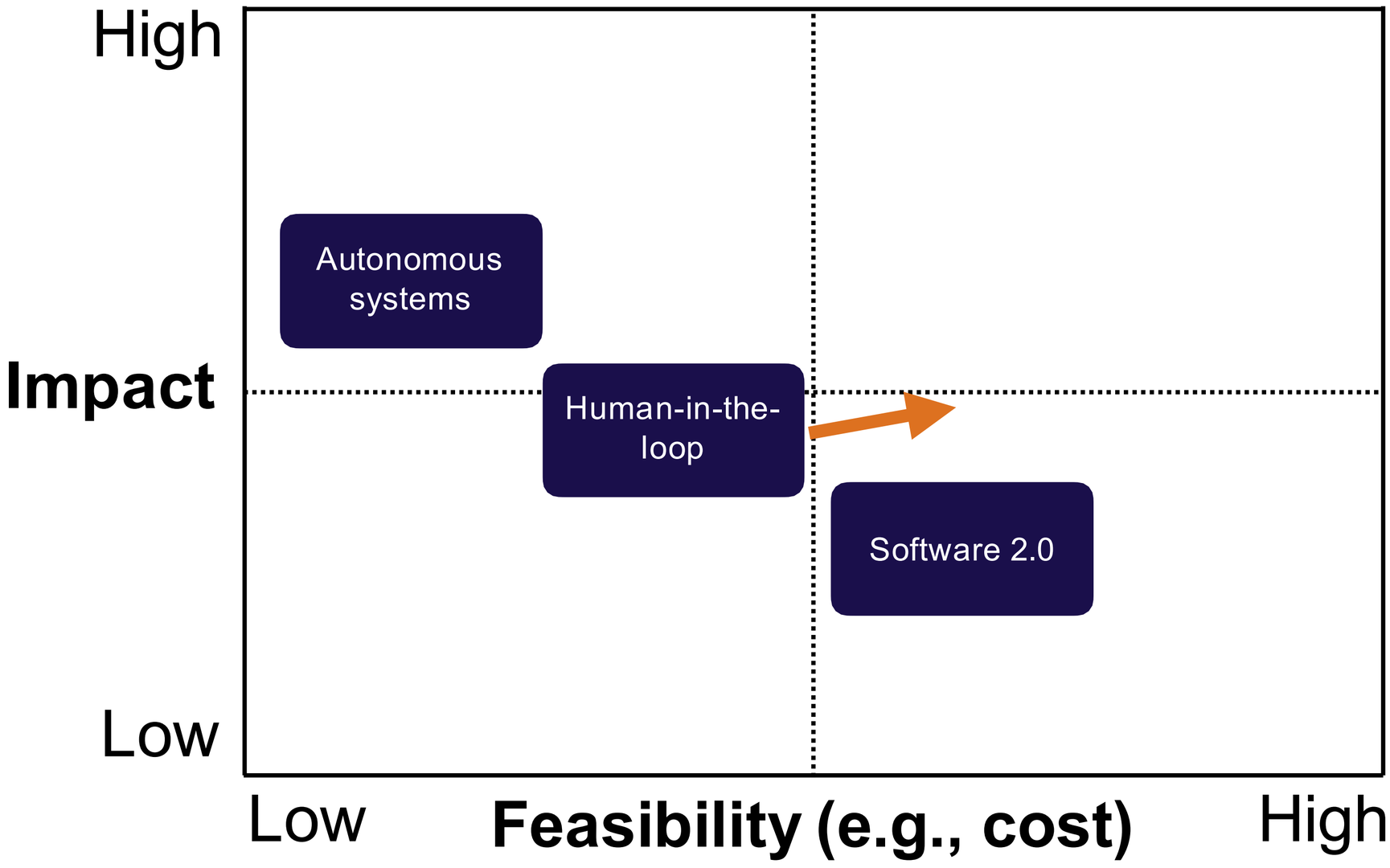


Machine learning product archetypes



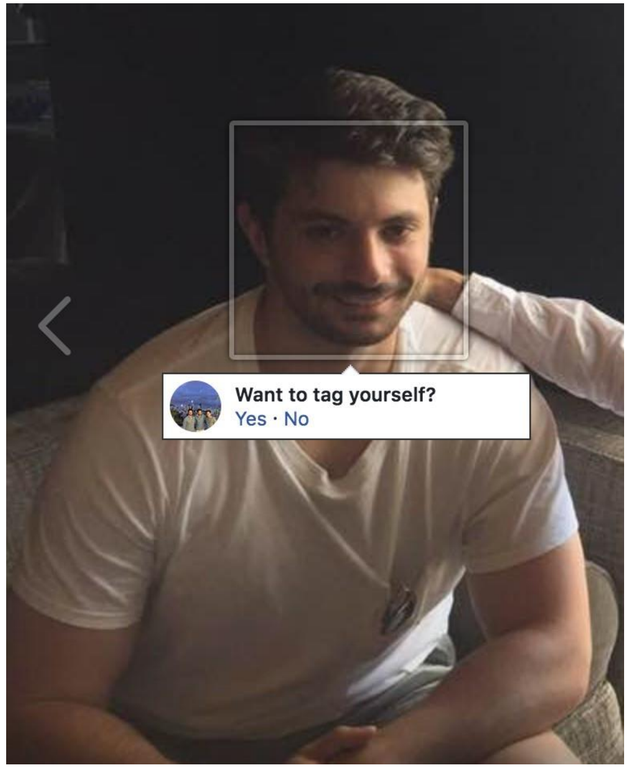
Implement a data loop that allows you to improve on this task and future ones

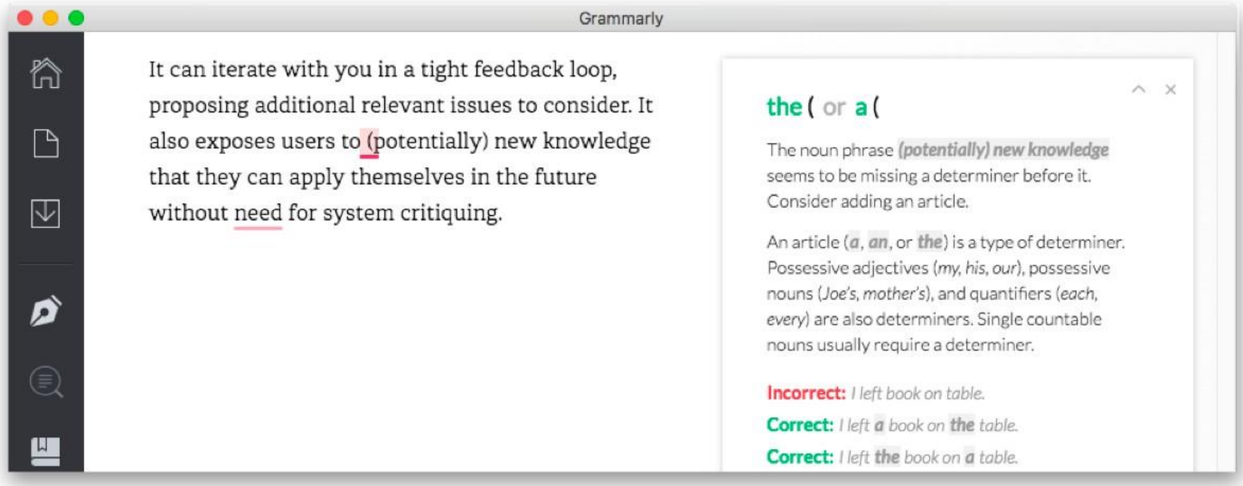
Machine learning product archetypes

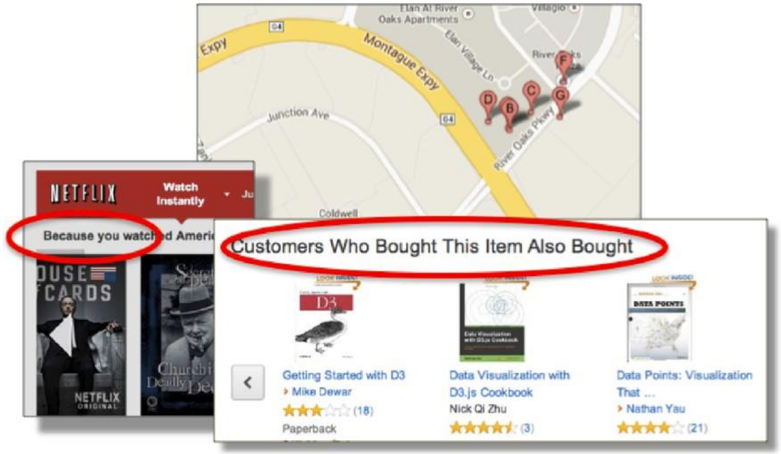


Good product design. Release a ‘good enough’ version

Product design can reduce need for accuracy







See "Designing Collaborative AI" (Ben Reinhardt and Belmer Negrillo):
https://medium.com/@Ben_Reinhardt/designing-collaborative-ai-5c1e8dbc8810

Apple’s ML product design guidelines

- What role does ML play in your app?
  - Critical or complementary?
  - Private or public?
  - Proactive or retroactive?
  - Visible or invisible?
  - Dynamic or static?

https://developer.apple.com/design/human-interface-guidelines/machine-learning/overview/introduction/

Apple’s ML product design guidelines

- How can you learn from your users?
  - Explicit feedback (e.g., “suggest less pop music”)
  - Implicit feedback (e.g., “like this song”)
  - Calibration during setup (e.g., scan your face for FaceID)
  - Corrections (e.g., fix mistakes model made)

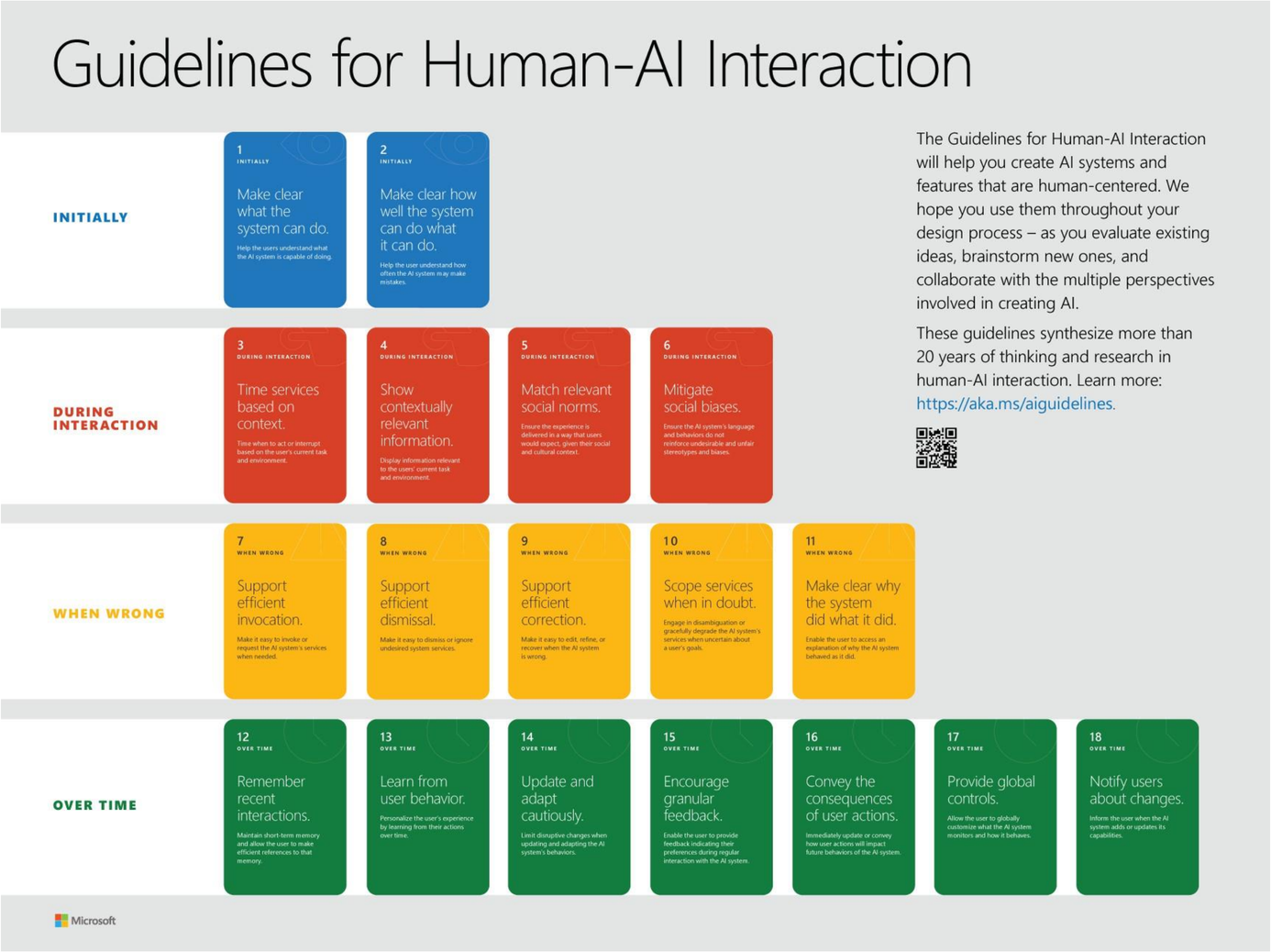
https://developer.apple.com/design/human-interface-guidelines/machine-learning/overview/introduction/

Apple’s ML product design guidelines

- How should your app handle mistakes?
  - Limitations (let your users know where you expect model to perform well)
  - Corrections (let your users succeed even if the model fails)
  - Attributions (help users understand where suggestions come from)
  - Confidence (help users gauge quality of results)

https://developer.apple.com/design/human-interface-guidelines/machine-learning/overview/introduction/

Machine learning product design



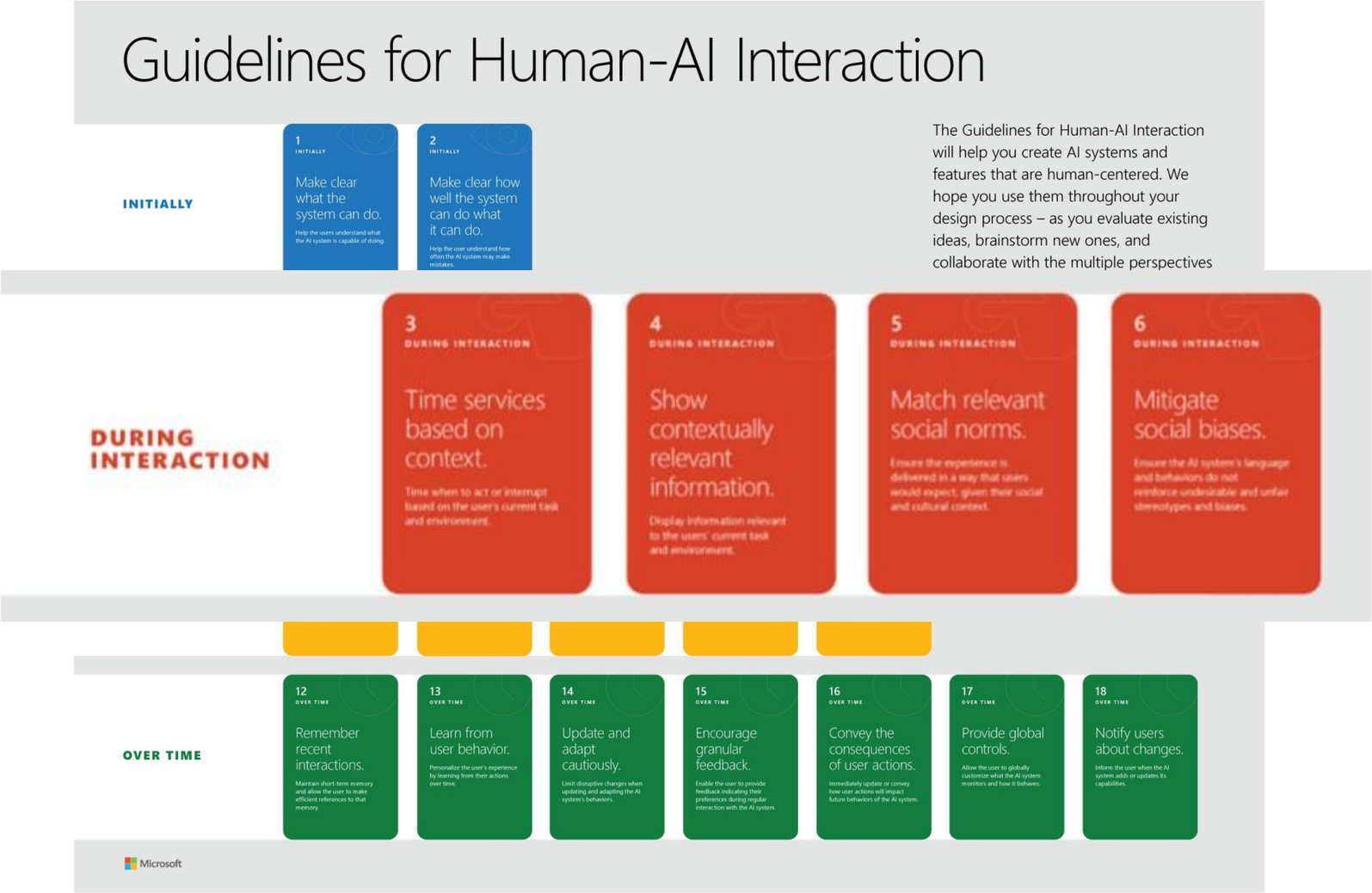
Guidelines for Human-AI Interaction
https://www.microsoft.com/en-us/research/project/guidelines-for-human-ai-interaction/

Machine learning product design

Guidelines for Human-AI Interaction

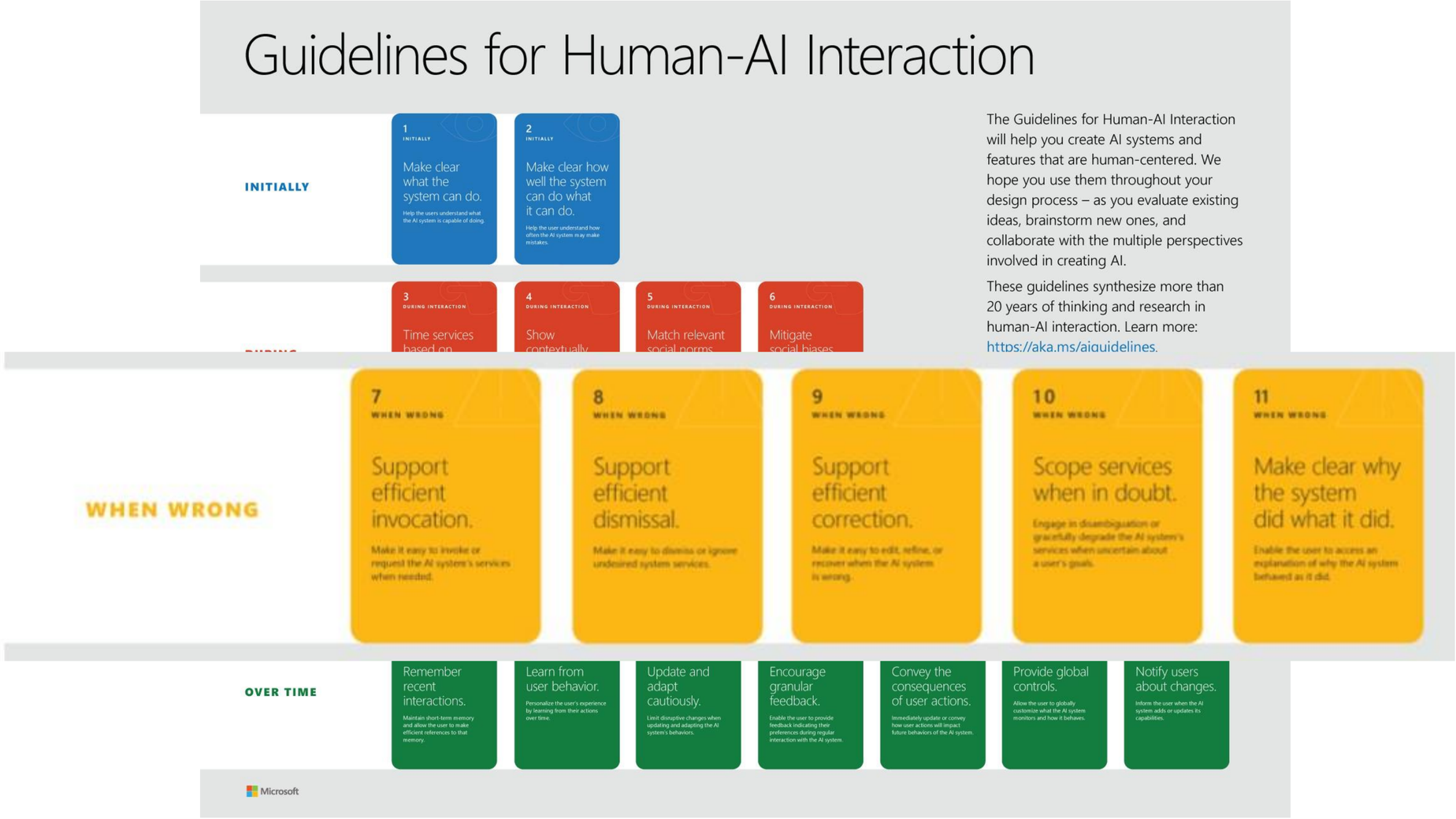
https://www.microsoft.com/en-us/research/project/guidelines-for-human-ai-interaction/

Machine learning product design



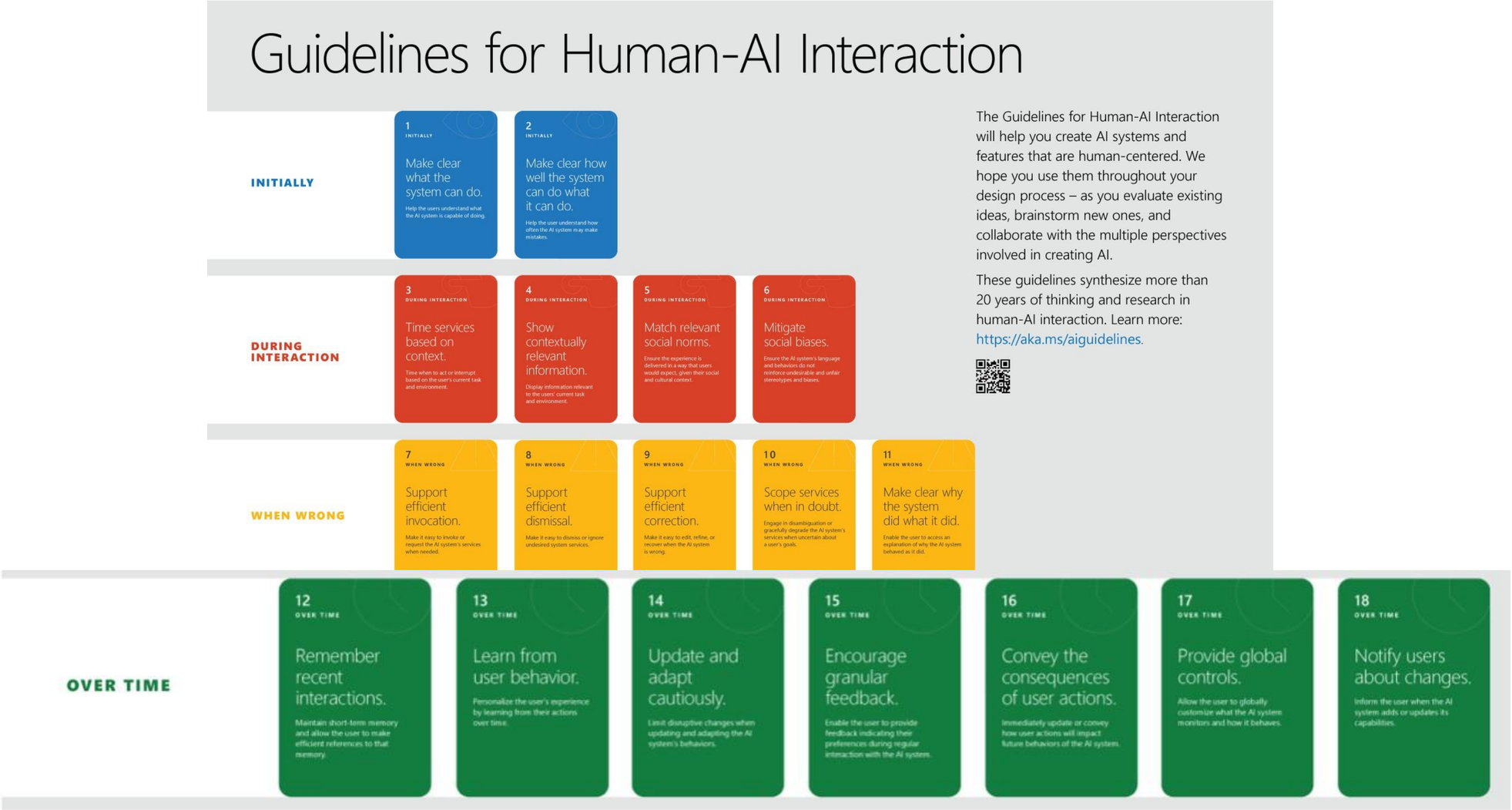
Guidelines for Human-AI Interaction
https://www.microsoft.com/en-us/research/project/guidelines-for-human-ai-interaction/

Machine learning product design



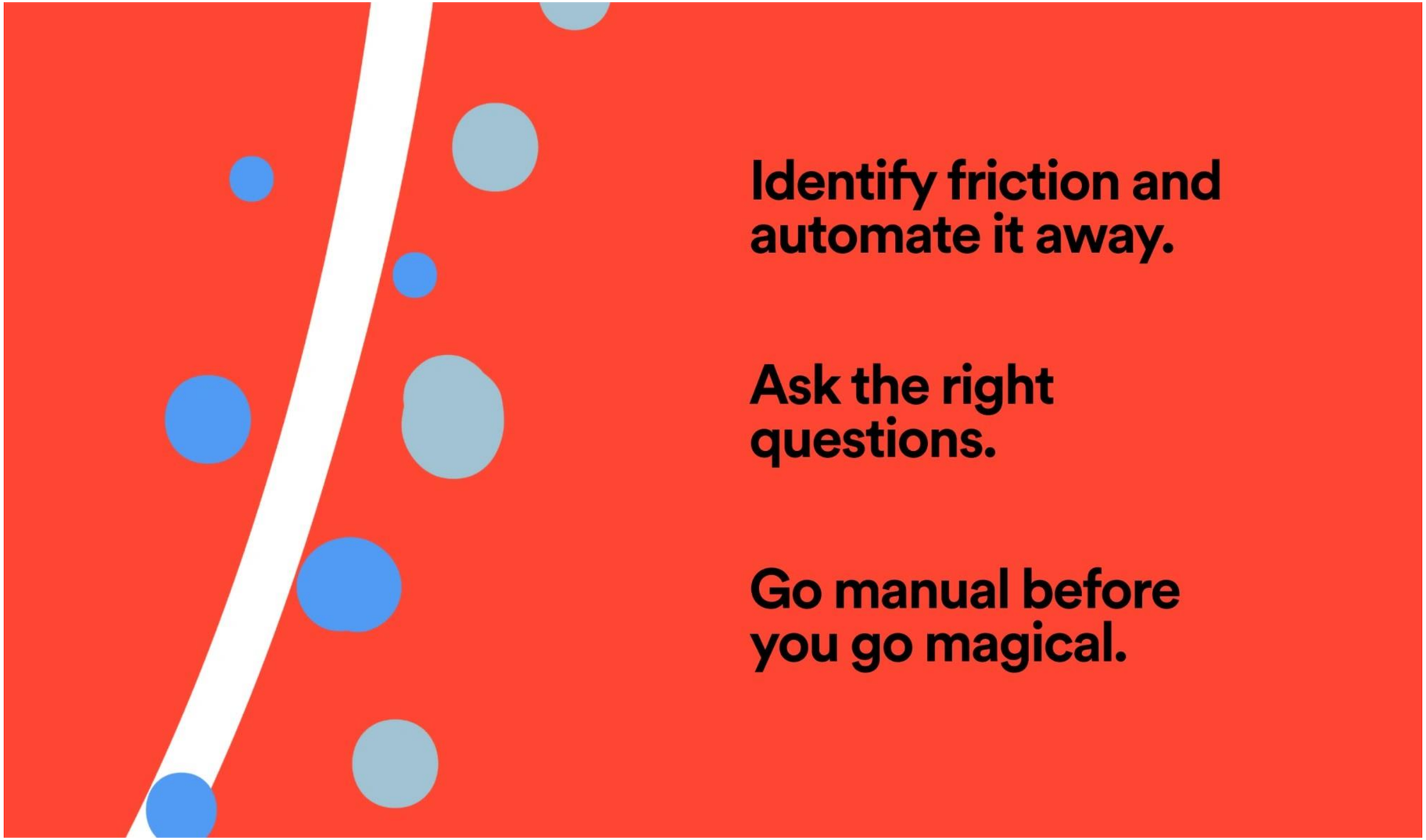
Guidelines for Human-AI Interaction
https://www.microsoft.com/en-us/research/project/guidelines-for-human-ai-interaction/

Machine learning product design



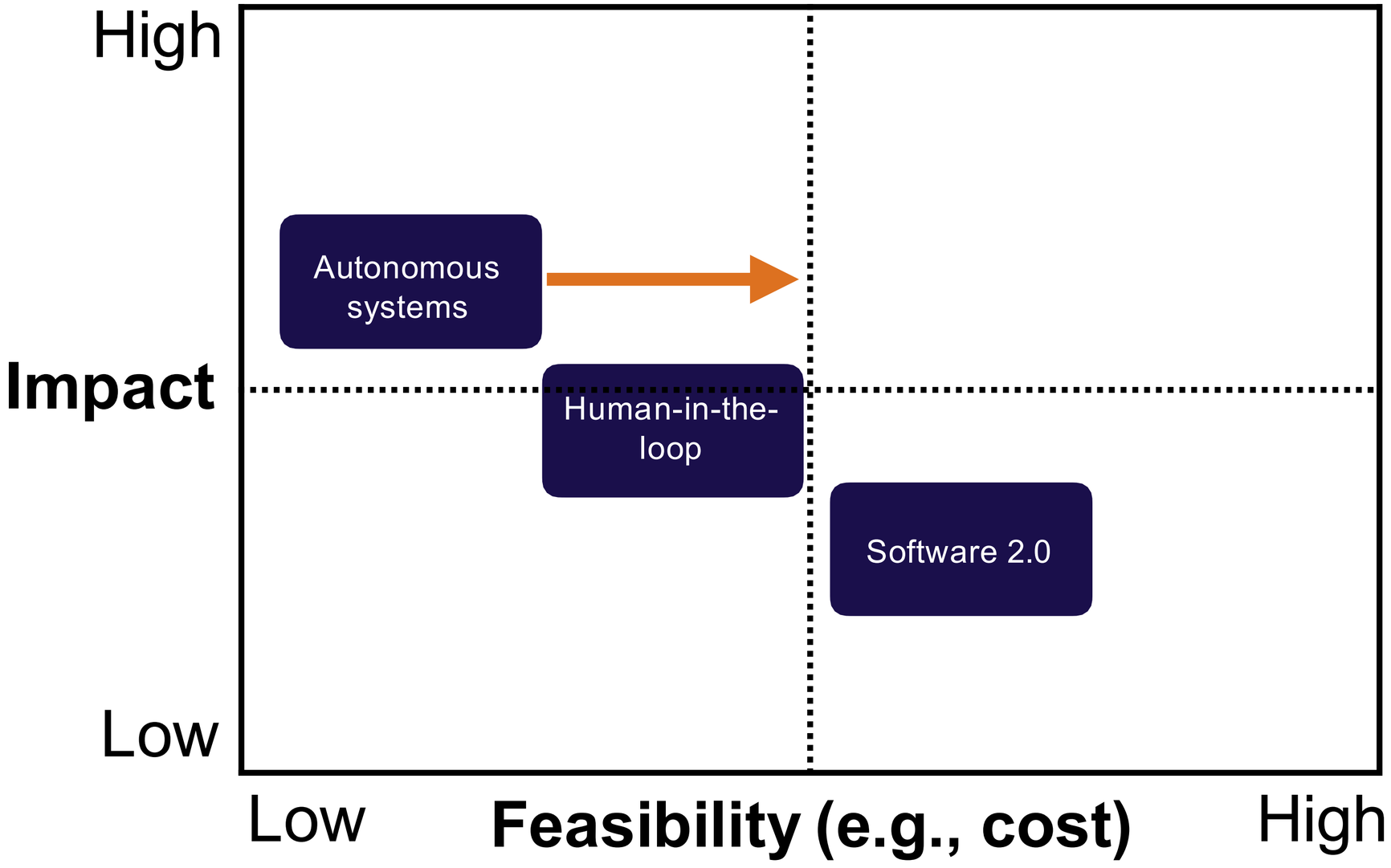
Guidelines for Human-AI Interaction
https://www.microsoft.com/en-us/research/project/guidelines-for-human-ai-interaction/

Product design can reduce need for accuracy



https://spotify.design/article/three-principles-for-designing-ml-powered-products

Machine learning product archetypes



Add humans in the loop. Add guardrails and/or limit initial scope.

Questions?

Module overview

- How to think about all of the activities in an ML project
- Assessing the feasibility and impact of your projects
- The main categories of ML projects, and the implications for project management

Metrics

- **How to pick a single number to optimize**
- How to know if your model is performing well

Key points for choosing a metric

A. The real world is messy; you usually care about lots of metrics

B. However, ML systems work best when optimizing a single number

C. As a result, you need to pick a formula for combining metrics

D. This formula can and will change!

Review of accuracy, precision, and recall

Confusion matrix

n=100	Predicted: NO	Predicted: YES	
Actual: NO	5	5	**10**
Actual: YES	45	45	**90**
	50	**50**	

Review of accuracy, precision, and recall

Confusion matrix

n=100	Predicted: NO	Predicted: YES	
Actual: NO	5	5	**10**
Actual: YES	45	45	**90**
	50	**50**	

$$\textbf{Accuracy} \quad \frac{\text{Correct}}{\text{Total}}$$

50%

Review of accuracy, precision, and recall

Confusion matrix

n=100	Predicted: NO	Predicted: YES	
Actual: NO	5	5	**10**
Actual: YES	45	45	**90**
	50	**50**	

Precision $\dfrac{\text{true positives}}{\text{true positives} + \text{false positives}}$

90%

Review of accuracy, precision, and recall

Confusion matrix

n=100	Predicted: NO	Predicted: YES	
Actual: NO	5	5	**10**
Actual: YES	45	45	**90**
	50	**50**	

Recall $\dfrac{\text{true positives}}{\text{Actual YES}}$

50%

Why choose a single metric?

	Precision	Recall
Model 1	**0.9**	**0.5**
Model 2	**0.8**	**0.7**
Model 3	**0.7**	**0.9**

Which is best?

How to combine metrics

- Simple average / weighted average

Combining precision and recall

	Precision	Recall
Model 1	**0.9**	**0.5**
Model 2	**0.8**	**0.7**
Model 3	**0.7**	**0.9**

Combining precision and recall

	Precision	Recall	(p + r) / 2
Model 1	**0.9**	**0.5**	**0.7**
Model 2	**0.8**	**0.7**	**0.75**
Model 3	**0.7**	**0.9**	**0.8**

Combining precision and recall

	Precision	Recall	(p + r) / 2
Model 1	**0.9**	**0.5**	**0.7**
Model 2	**0.8**	**0.7**	**0.75**
Model 3	**0.7**	**0.9**	**0.8**

How to combine metrics

- Simple average / weighted average

How to combine metrics

- Simple average / weighted average
- Threshold n-1 metrics, evaluate the nth

Thresholding metrics

Choosing which metrics to threshold

- Domain judgment (e.g., which metrics can you engineer around?)
- Which metrics are least sensitive to model choice?
- Which metrics are closest to desirable values?

Choosing threshold values

- Domain judgment (e.g., what is an acceptable tolerance downstream? What performance is achievable?)
- How well does the baseline model do?
- How important is this metric right now?

Combining precision and recall

	Precision	Recall	(p + r) / 2
Model 1	**0.9**	**0.5**	**0.7**
Model 2	**0.8**	**0.7**	**0.75**
Model 3	**0.7**	**0.9**	**0.8**

Combining precision and recall

	Precision	Recall	(p + r) / 2	p @(r > 0.6)
Model 1	0.9	0.5	0.7	0.0
Model 2	0.8	0.7	0.75	0.8
Model 3	0.7	0.9	0.8	0.7

Combining precision and recall

	Precision	Recall	(p + r) / 2	p @(r > 0.6)
Model 1	0.9	0.5	0.7	0.0
Model 2	0.8	0.7	0.75	0.8
Model 3	0.7	0.9	0.8	0.7

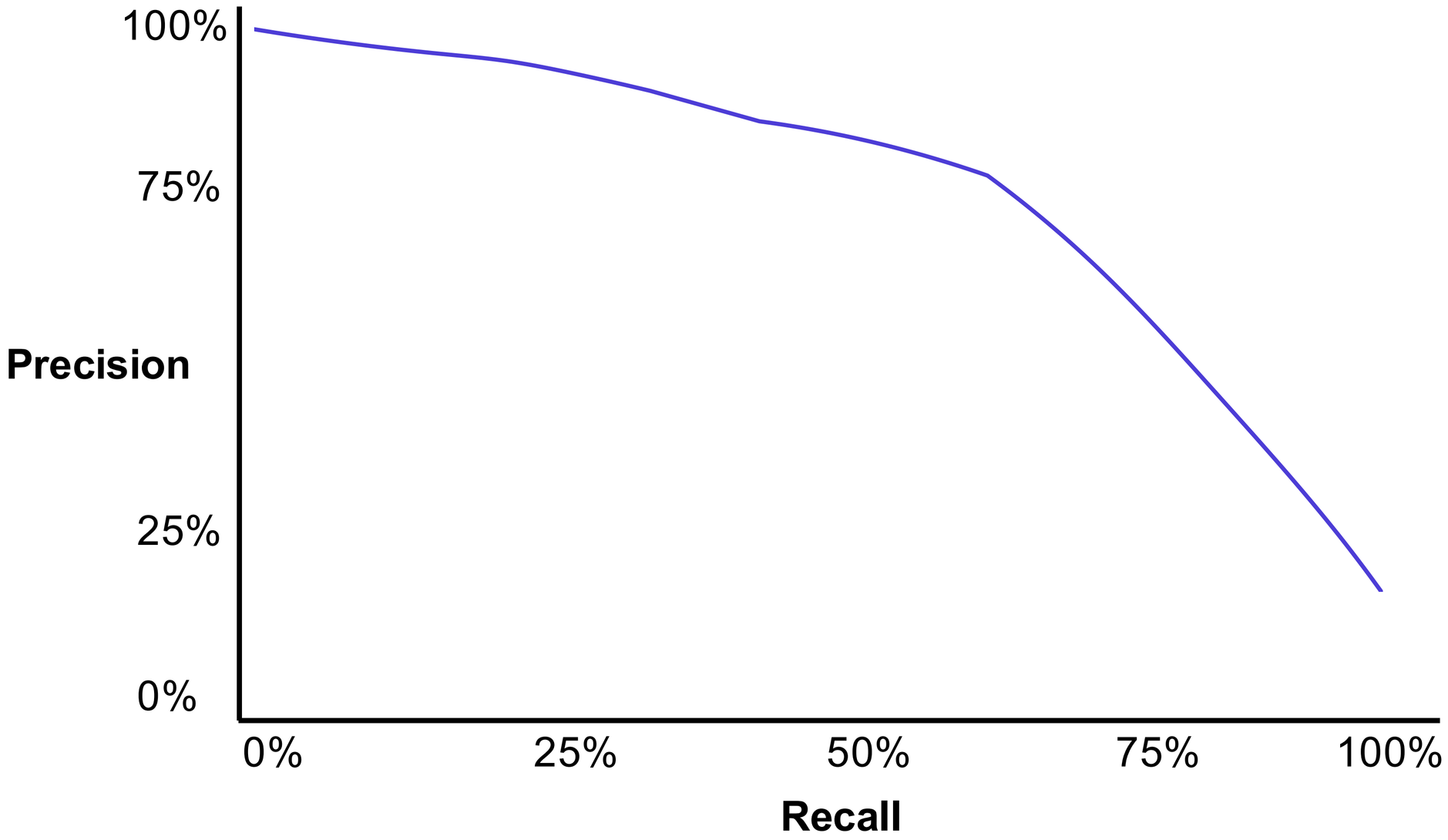
How to combine metrics

- Simple average / weighted average
- Threshold n-1 metrics, evaluate the nth

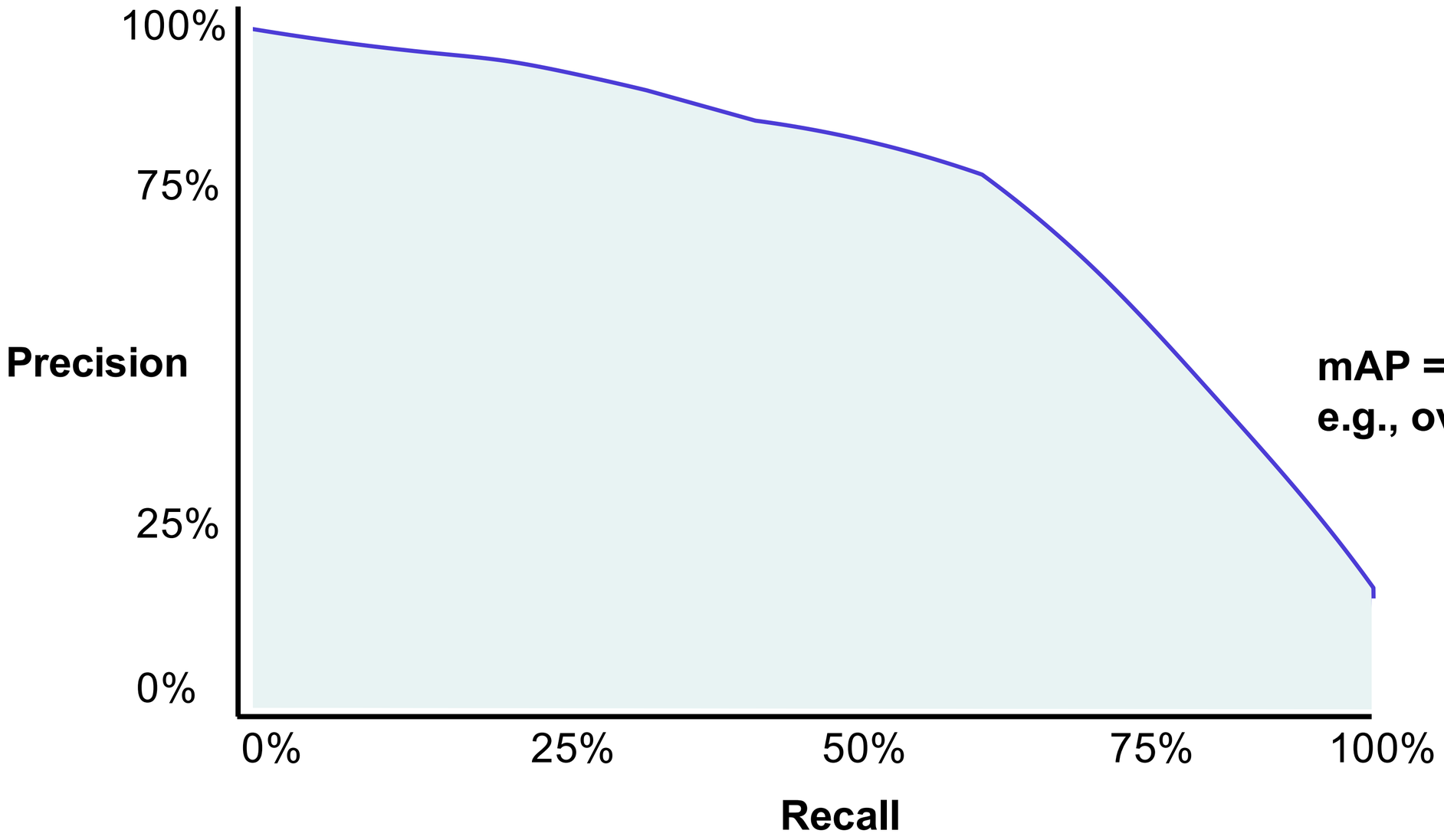
How to combine metrics

- Simple average / weighted average
- Threshold n-1 metrics, evaluate the nth
- More complex / domain-specific formula

Domain-specific metrics: mAP



Domain-specific metrics: mAP



mAP = *mean* AP,
e.g., over classes

Combining precision and recall

	Precision	Recall	(p + r) / 2	p @(r > 0.6)
Model 1	0.9	0.5	0.7	0.0
Model 2	0.8	0.7	0.75	0.8
Model 3	0.7	0.9	0.8	0.7

Combining precision and recall

	Precision	Recall	(p + r) / 2	p @(r > 0.6)	mAP
Model 1	0.9	0.5	0.7	0.0	0.7
Model 2	0.8	0.7	0.75	0.8	0.6
Model 3	0.7	0.9	0.8	0.7	0.6

Combining precision and recall

	Precision	Recall	(p + r) / 2	p @(r > 0.6)	mAP
Model 1	**0.9**	**0.5**	**0.7**	**0.0**	**0.7**
Model 2	**0.8**	**0.7**	**0.75**	**0.8**	**0.6**
Model 3	**0.7**	**0.9**	**0.8**	**0.7**	**0.6**

Example: choosing a metric for pose estimation

Xiang, Yu, et al. "PoseCNN: A Convolutional Neural Network for 6D Object Pose Estimation in Cluttered Scenes." arXiv preprint arXiv:1711.00199 (2017).

(x, y, z) **Position (L2 loss)**

$(¢, ✓, \ )$ **Orientation (L2 loss)**

t **Prediction time**

Example: choosing a metric for pose estimation

- **Enumerate requirements**
  - Downstream goal is real-time robotic grasping
  - Position error must be <1cm, not sure exactly how precise is needed
  - Angular error <5 degrees
  - Must run in 100ms to work in real-time

Example: choosing a metric for pose estimation

- Enumerate requirements
- **Evaluate current performance**
  - Train a few models

Example: choosing a metric for pose estimation

- Enumerate requirements
- Evaluate current performance
- **Compare current performance to requirements**
  - Position error between 0.75 and 1.25cm (depending on hyperparameters)
  - All angular errors around 60 degrees
  - Inference time ~300ms

Example: choosing a metric for pose estimation

- Enumerate requirements
- Evaluate current performance
- **Compare current performance to requirements**
  - Prioritize angular error
  - Threshold position error at 1cm
  - Ignore run time for now

Example: choosing a metric for pose estimation

- Enumerate requirements
- Evaluate current performance
- Compare current performance to requirements
- **Revisit metric as your numbers improve**

Key points for choosing a metric

A. The real world is messy; you usually care about lots of metrics

B. However, ML systems work best when optimizing a single number

C. As a result, you need to pick a formula for combining metrics

D. This formula can and will change!

Questions?

Module overview

- How to think about all of the activities in an ML project
- Assessing the feasibility and impact of your projects
- The main categories of ML projects, and the implications for project management
- How to pick a single number to optimize
- **How to know if your model is performing well**

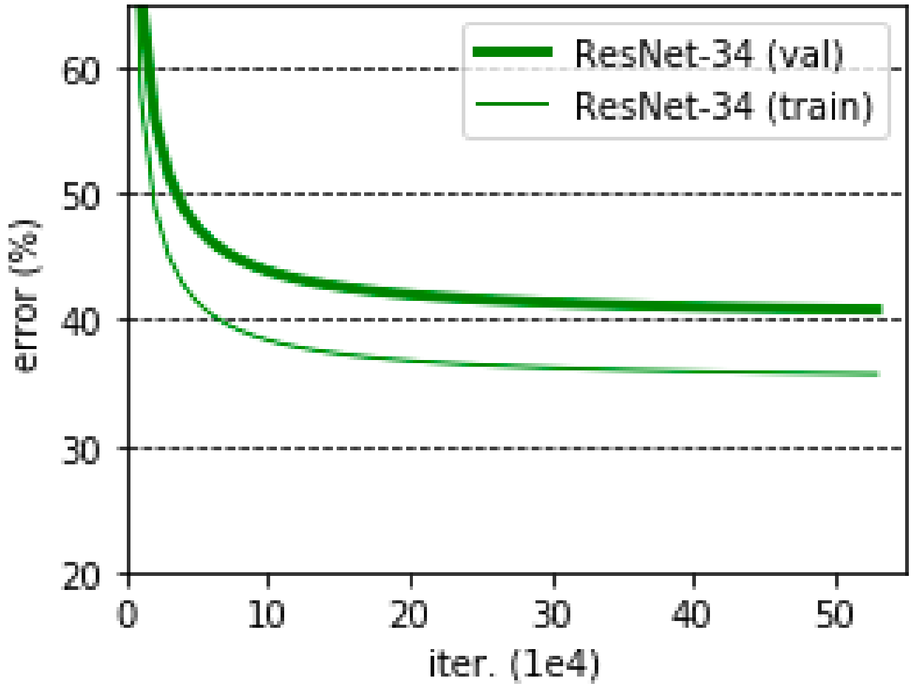
Baselines

Key points for choosing baselines

A. Baselines give you a lower bound on expected model performance

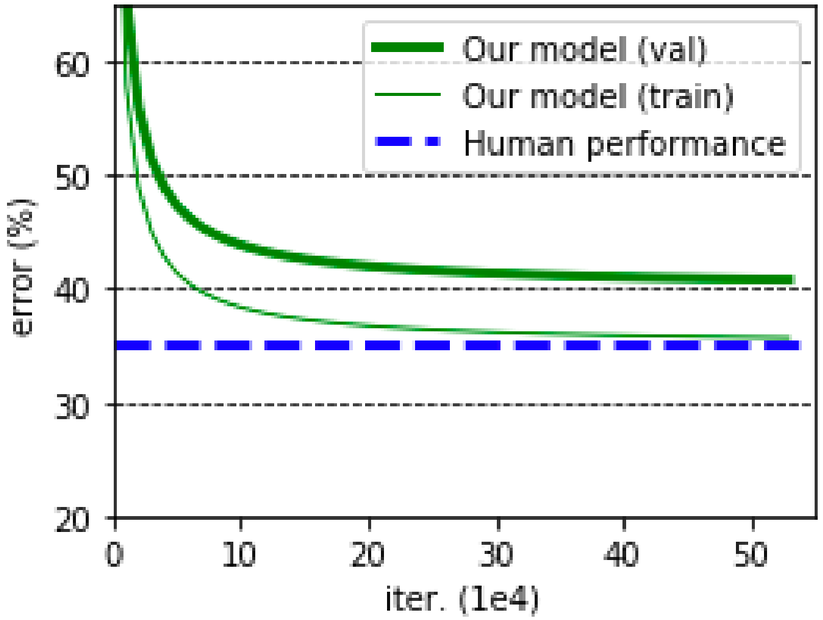
B. The tighter the lower bound, the more useful the baseline (e.g., published results, carefully tuned pipelines, & human baselines are better)

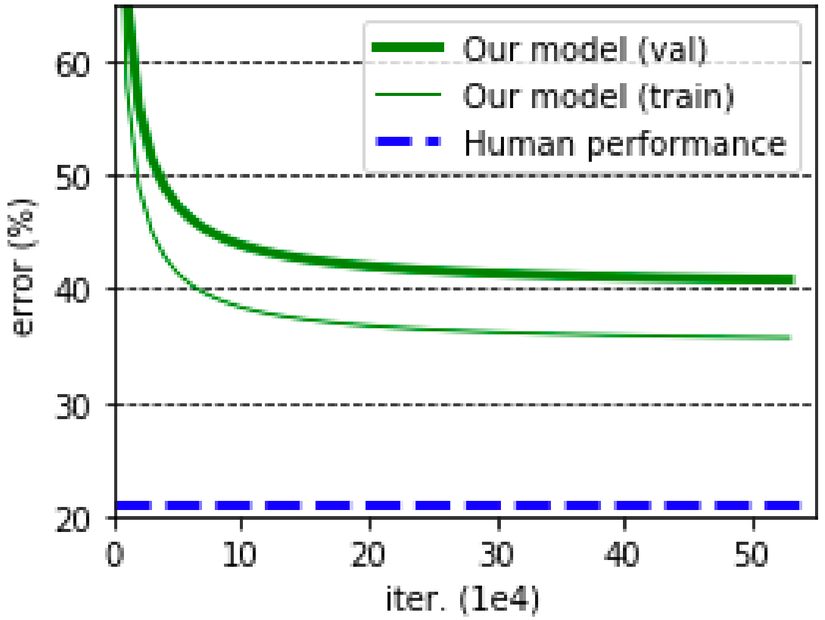
Why are baselines important?



Why are baselines important?

Same model, different baseline —> different next steps





Where to look for baselines

External baselines

- Business / engineering requirements

Where to look for baselines

External baselines

- Business / engineering requirements
- Published results

$\vdash\!\!\rightarrow$ **Make sure comparison is fair!**

Where to look for baselines

External baselines

- Business / engineering requirements
- Published results

Internal baselines

- Scripted baselines (e.g., OpenCV, rules-based)

Where to look for baselines

External baselines

- Business / engineering requirements
- Published results

Internal baselines

- Scripted baselines (e.g., OpenCV, rules-based)
- Simple ML baselines (e.g., bag of words, linear regression)

Where to look for baselines

External baselines

- Business / engineering requirements
- Published results

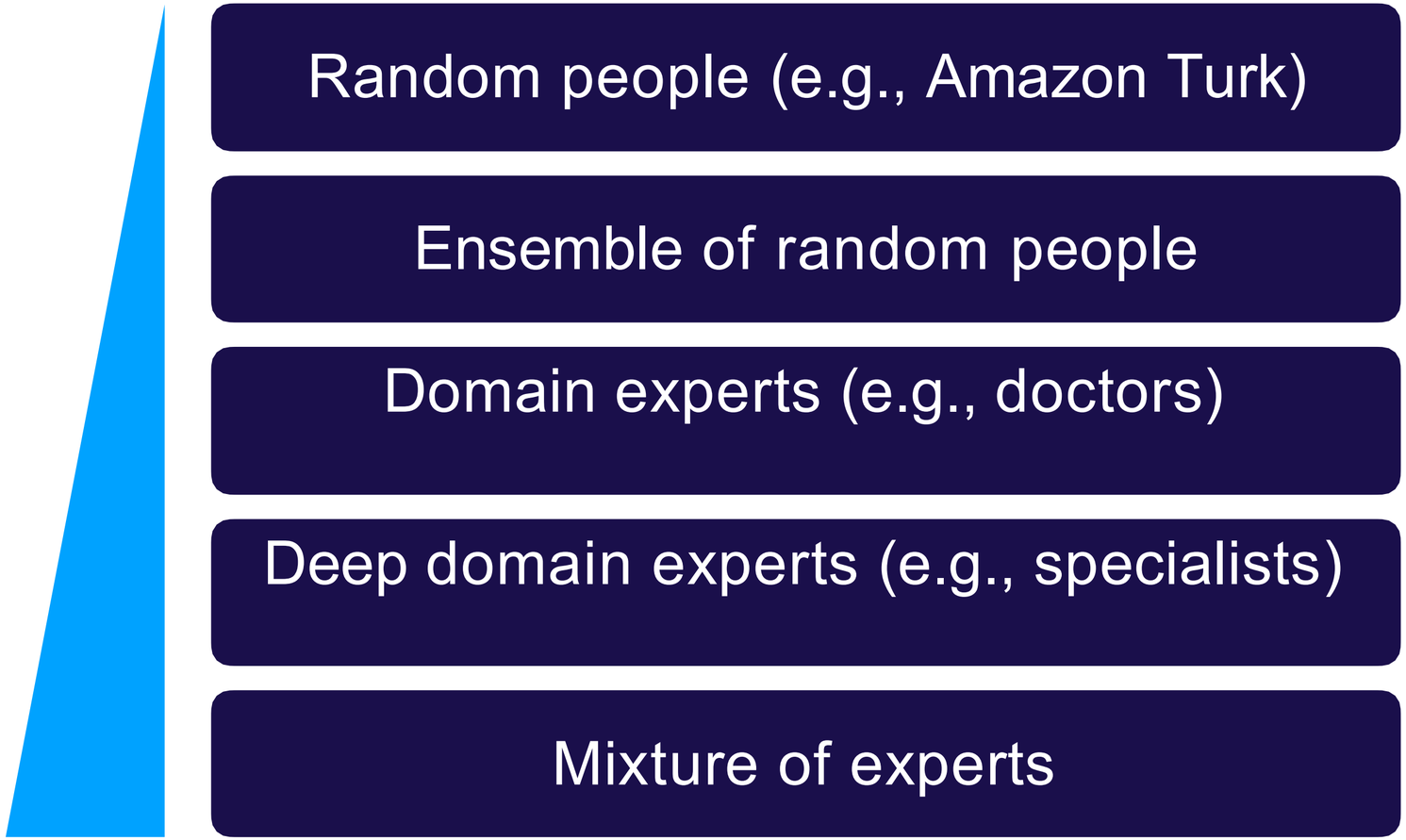
Internal baselines

- Scripted baselines (e.g., OpenCV, rules-based)
- Simple ML baselines (e.g., bag of words, linear regression)
- Human performance

How to create good human baselines

Quality of baseline

Low



Ease of data collection

High

Low

How to create good human baselines

- Highest quality that allows more data to be labeled easily
- More specialized domains need more skilled labelers
- Find cases where model performs worse and concentrate data collection there

More on labeling in data lecture!

Key points for choosing baselines

A. Baselines give you a lower bound on expected model performance

B. The tighter the lower bound, the more useful the baseline (e.g., published results, carefully tuned pipelines, human baselines are better)

Questions?

Conclusion

Lifecycle

- **ML projects are iterative. Deploy something fast to begin the cycle.**

Prioritizing projects

- **Choose projects that that are high impact with low cost of wrong predictions**

Archetypes

- **The secret sauce to making projects work well is to build automated data flywheels**

Metrics

- **In the real world you care about many things, but you should always have just one you're working on**

Baselines

- **Good baselines help you invest your effort the right way**

Where to go to learn more

- Andrew Ng's "Machine Learning Yearning"
- Andrej Karpathy's "Software 2.0"
- Agrawal's "The Economics of AI"
- Chip Huyen's "Introdution to Machine Learning Systems Design"
- Apple's "Human Interface Guidelines for Machine Learning"
- Google's "Rules of Machine Learning"

Thank you!